

Coherent-Constructible Correspondence

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Abstract

This note gives an expository account of the toric coherent–constructible correspondence, mainly based on [FLTZ09]. The main focus is the sheaf-theoretic part of the correspondence: on one side, coherent sheaves on a toric variety; on the other side, constructible sheaves on a real vector space determined by the fan. We explain how the combinatorial data of a toric variety controls both sides and how the correspondence is proved by comparing their natural families of generators.

Besides the sheaf-theoretic proof, this note also includes two supplementary parts. Following the discussion in [FLTZ09], we briefly recall the Fukaya-categorical interpretation of the toric CCC, explaining how the constructible side may be viewed through microlocal sheaf theory and Fukaya categories. In the final part, based mainly on [FLTZ14, Kuw20, FLTZ11, HS23], we give a short overview of several related generalizations, including toric stacks, toric fibrations, and Fourier–Mukai-type constructions.

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1 Introduction

1.1 Main Aim and Strategy

The coherent–constructible correspondence is a bridge between two different types of sheaf theories. On one side, one studies equivariant coherent sheaves on a toric variety. On the other side, one studies constructible sheaves on a real vector space, with singular support restricted by a conic subset determined by the fan. The main point of the correspondence is that the algebraic and equivariant data of a toric variety can be translated into polyhedral and microlocal data on the constructible side.

The main goal of this paper is to explain and prove the coherent–constructible correspondence for smooth toric varieties. More precisely, if X_Σ is a smooth toric variety with acting torus T , character lattice M , and fan $\Sigma \subset N_\mathbb{R}$, then the correspondence takes the form of a quasi-equivalence of dg categories

$$\kappa_\Sigma : \text{Perf}_T(X_\Sigma) \xrightarrow{\sim} Sh_{\Lambda_\Sigma}(M_\mathbb{R}).$$

Here $\text{Perf}_T(X_\Sigma)$ is the dg category of T -equivariant perfect complexes on X_Σ , and $Sh_{\Lambda_\Sigma}(M_\mathbb{R})$ is the dg category of constructible sheaves on $M_\mathbb{R}$ whose singular support is contained in the FLTZ skeleton Λ_Σ . The FLTZ skeleton is a conic subset of $T^*M_\mathbb{R}$ defined directly from the fan Σ .

The proof follows the standard generator strategy for dg categories. The key point is that, in the toric setting, both sides admit explicit families of generators: the coherent theta objects on the coherent side and the constructible theta sheaves on the constructible side. These generators are indexed by the same combinatorial data coming from the fan. Therefore, the equivalence is not proved by comparing arbitrary objects directly. Instead, it is reduced to comparing the Hom-complexes between the two families of theta generators. Once these Hom-complexes are identified and the generation results are established on both sides, the coherent–constructible correspondence follows from the generator criterion.

1.2 Organization of the Note

The organization of the paper is as follows. Section 2 recalls the symplectic and Kähler background of toric varieties, especially the relation between ample toric line bundles and moment polytopes. Section 3 fixes the categorical and sheaf-theoretic conventions used later, including dg categories, equivariant coherent sheaves, constructible sheaves, and singular support.

Sections 4, 5, and 6 form the main part of the paper. Section 4 describes the coherent side and introduces the coherent theta generators. Section 5 constructs the constructible side through the FLTZ skeleton and the constructible theta sheaves. Section 6 proves the toric coherent–constructible correspondence by comparing the Hom-complexes between these generators.

The remaining sections are supplementary. Section 7 briefly discusses the Fukaya-categorical interpretation of the constructible side, following the perspective of [FLTZ09]. Section 8 presents several explicit examples of the toric CCC; these computations are based on the general framework of [FLTZ09]. Section 9 gives a short overview of related generalizations: toric Deligne–Mumford stacks as in [FLTZ14], nonequivariant toric stacks as in [Kuw20], Fourier–Mukai-type constructions as in [FLTZ11], and toric fibrations as in [HS23].

Readers who are already familiar with dg categories, constructible sheaves, singular support, and the basic symplectic background of toric varieties may start directly from Section 4.

1.3 Standard Notation

We use the standard notation of toric geometry. Let T be an algebraic torus over $\mathbb{C}$. We write

$$N = \text{Hom}(\mathbb{C}^*, T), \quad M = \text{Hom}(T, \mathbb{C}^*)$$

for the cocharacter and character lattices. Thus $M = \text{Hom}_\mathbb{Z}(N, \mathbb{Z})$. Their realifications are denoted by

$$N_\mathbb{R} = N \otimes_\mathbb{Z} \mathbb{R}, \quad M_\mathbb{R} = M \otimes_\mathbb{Z} \mathbb{R}.$$

For a fan $\Sigma \subset N_{\mathbb{R}}$, we denote by X_{Σ} the associated toric variety. If $\sigma \in \Sigma$ is a cone, then its dual cone is

$$\sigma^{\vee} = \{m \in M_{\mathbb{R}} \mid \langle m, n \rangle \geq 0 \text{ for all } n \in \sigma\}.$$

The corresponding affine toric chart is written as

$$U_{\sigma} = \text{Spec } \mathbb{C}[\sigma^{\vee} \cap M],$$

and its inclusion into X_{Σ} is denoted by

$$i_{\sigma} : U_{\sigma} \hookrightarrow X_{\Sigma}.$$

For a cone σ , we write

$$\sigma^{\vee} = \{m \in M_{\mathbb{R}} \mid \langle m, n \rangle \geq 0 \text{ for all } n \in \sigma\}$$

for its dual cone, and

$$\sigma^{\perp} = \{m \in M_{\mathbb{R}} \mid \langle m, n \rangle = 0 \text{ for all } n \in \sigma\}$$

for its annihilator. We also use

$$M_{\sigma} = M/(\sigma^{\perp} \cap M).$$

2 Symplectic Aspects of Toric Varieties

In this section only, we restrict attention to smooth projective toric varieties. This projectivity assumption is not part of the sheaf-theoretic proof of the toric coherent–constructible correspondence given later. It is imposed here because the symplectic discussion uses ample toric line bundles, Kähler forms, moment maps, and moment polytopes, which are most naturally formulated in the projective setting.

Let X be a smooth projective toric variety. Since X is projective, its associated complex manifold admits a Kähler form. Hence X may also be viewed as a symplectic manifold. The purpose of this section is to recall the basic symplectic notions needed for this viewpoint and to explain, at the level of definitions, how such a symplectic structure can be obtained from the algebraic data of an ample toric line bundle.

Throughout this section, we use both the Zariski topology and the usual analytic topology when needed. We regard X both as an algebraic variety and as its associated complex manifold; when no confusion can arise, we will not distinguish these viewpoints explicitly. To keep the exposition concise, we state the standard facts used in this section without proof.

2.1 Symplectic Manifolds

Let X be a smooth real manifold. A **symplectic form** on X is a closed non-degenerate two-form $\omega \in \Omega^2(X)$. Thus $d\omega = 0$, and for every point $x \in X$, the bilinear form $\omega_x : T_x X \times T_x X \rightarrow \mathbb{R}$ is non-degenerate. Equivalently, the map $T_x X \rightarrow T_x^* X$, $v \mapsto \omega_x(v, -)$, is an isomorphism for every $x \in X$. A pair (X, ω) is called a **symplectic manifold**.

A diffeomorphism $\phi : X \rightarrow X$ is called a **symplectomorphism** if it preserves the symplectic form, namely if $\phi^* \omega = \omega$. Thus a symplectomorphism is an automorphism of the symplectic manifold (X, ω) .

Let G be a Lie group acting smoothly on X . We say that the action is a **symplectic action** if every element $g \in G$ acts by a symplectomorphism, equivalently if $g^* \omega = \omega$ for all $g \in G$. We denote the action by $\alpha : G \times X \rightarrow X$, $(g, x) \mapsto g \cdot x$. For each $x \in X$, the action determines an orbit map $\alpha_x : G \rightarrow X$, $g \mapsto g \cdot x$.

Let $\mathfrak{g}$ be the Lie algebra of G . For $\xi \in \mathfrak{g}$, the corresponding one-parameter subgroup is $t \mapsto \exp(t\xi)$. Differentiating the orbit map along this one-parameter subgroup gives a tangent vector at x :

$$\xi_X(x) = \left. \frac{d}{dt} \right|_{t=0} \exp(t\xi) \cdot x \in T_x X.$$

As x varies, this defines a vector field $\xi_X \in \Gamma(TX)$. The assignment $\mathfrak{g} \rightarrow \Gamma(TX)$, $\xi \mapsto \xi_X$, is called the **infinitesimal action** of G on X . The vector field ξ_X is called the **fundamental vector field** generated by ξ .

A symplectic G -action is called **Hamiltonian** if there exists a smooth map $\mu : X \rightarrow \mathfrak{g}^*$ such that, for every $\xi \in \mathfrak{g}$,

$$d\langle \mu, \xi \rangle = \iota_{\xi_X} \omega.$$

Here $\langle \mu, \xi \rangle : X \rightarrow \mathbb{R}$ is the function obtained by pairing $\mu(x) \in \mathfrak{g}^*$ with ξ , and $\iota_{\xi_X} \omega$ denotes contraction of ω with the vector field ξ_X . The map μ is called a **moment map**, and the triple (X, ω, μ) is called a **Hamiltonian G -space**.

The moment map is not unique in general. If $\mu : X \rightarrow \mathfrak{g}^*$ is a moment map, then $\mu + c$ is also a moment map for any constant element $c \in \mathfrak{g}^*$. Thus a moment map is determined only up to addition of a constant element of $\mathfrak{g}^*$.

A moment map is called **G -equivariant** if $\mu(g \cdot x) = \text{Ad}_g^* \mu(x)$ for all $g \in G$ and $x \in X$. Here Ad^* is the coadjoint action, namely the dual action to the adjoint representation $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$.

2.2 Kähler Manifolds and Complex Structure

Let X be a smooth manifold, and let $D \subset TX$ be a smooth subbundle of the tangent bundle. We call D a **distribution** on X . The distribution D is called **integrable** if, for every point $x \in X$, there exists an immersed submanifold $Y \subset X$ passing through x such that $T_y Y = D_y$ for every $y \in Y$. Such a submanifold Y is called an **integral manifold** of D .

An **almost complex structure** on X is a bundle endomorphism $J : TX \rightarrow TX$ such that $J^2 = -\text{id}_{TX}$. If J is integrable, then X is a complex manifold, and J is called its **complex structure**.

Suppose now that X is a complex manifold with complex structure J . A Riemannian metric g on X is called **Hermitian** if it is compatible with J , namely if $g(Jv, Jw) = g(v, w)$ for all tangent vectors v, w . Associated to a Hermitian metric is the real two-form ω defined by $\omega(v, w) = g(Jv, w)$. This form is called the **fundamental form** of the Hermitian metric.

The Hermitian metric g is called **Kähler** if $d\omega = 0$. In this case ω is called a **Kähler form**, and the triple (X, J, ω) is called a **Kähler manifold**.

A Kähler form is in particular a symplectic form: it is a closed non-degenerate real two-form. Thus every Kähler manifold is naturally a symplectic manifold. Conversely, on a complex manifold, a symplectic form compatible with the complex structure determines a Kähler metric by $g(v, w) = \omega(v, Jw)$.

2.3 Hermitian Line Bundles and Curvature

We recall a standard construction from complex differential geometry. Let X be a complex manifold, and let L be a holomorphic line bundle. A **Hermitian metric** on L is a smoothly varying Hermitian inner product $h_x : L_x \times L_x \rightarrow \mathbb{C}$ on the fibers of L . If s is a local non-vanishing holomorphic frame of L , then the metric is determined locally by the positive smooth function $h(s, s) > 0$.

There is a unique connection ∇_h on L , called the **Chern connection**, which is compatible with both the holomorphic structure and the Hermitian metric. Compatibility with the holomorphic structure means that $\nabla_h^{0,1} = \bar{\partial}_L$. Compatibility with the metric means that, for smooth local sections s_1, s_2 of L ,

$$dh(s_1, s_2) = h(\nabla_h s_1, s_2) + h(s_1, \nabla_h s_2).$$

The **curvature** of the Chern connection is $F_h = \nabla_h^2$. Since L is a line bundle, F_h may be regarded as a complex-valued two-form on X .

Locally, if s is a holomorphic frame and $h(s, s) = e^{-\varphi}$, then the curvature is expressed in terms of the local potential φ . With the usual convention, the associated real $(1, 1)$ -form is

$$\omega_h = \sqrt{-1} \partial \bar{\partial} \varphi.$$

Equivalently, up to the chosen sign convention, one may write $\omega_h = \sqrt{-1} F_h$.

If L is positive, then ω_h is a Kähler form. Thus a positive Hermitian metric on a positive holomorphic line bundle produces a symplectic form on the underlying smooth manifold.

2.4 Algebraic Tori and Compact Real Tori

Let X be a smooth projective toric variety of complex dimension n . Its acting algebraic torus T is a complex algebraic group. Under an identification $T \cong (\mathbb{C}^*)^n$, it contains the maximal compact real subtorus $T_{\mathbb{R}} \cong (S^1)^n$, explicitly

$$T_{\mathbb{R}} = \{(z_1, \dots, z_n) \in (\mathbb{C}^*)^n \mid |z_1| = \dots = |z_n| = 1\}.$$

The **compact torus action** $T_{\mathbb{R}} \times X \rightarrow X$ is obtained by restricting the algebraic T -action on X to the subgroup $T_{\mathbb{R}} \subset T$.

The real vector space $N_{\mathbb{R}}$ is naturally identified with the Lie algebra of the compact torus, so $\text{Lie}(T_{\mathbb{R}}) \cong N_{\mathbb{R}}$. Dually, one has $\text{Lie}(T_{\mathbb{R}})^* \cong M_{\mathbb{R}}$. Under these identifications, the exponential map is

$$\exp : \text{Lie}(T_{\mathbb{R}}) \cong N_{\mathbb{R}} \longrightarrow T_{\mathbb{R}},$$

and its kernel is the lattice N . Hence $T_{\mathbb{R}} \cong N_{\mathbb{R}}/N$. Similarly, the dual compact torus is $T_{\mathbb{R}}^{\vee} \cong M_{\mathbb{R}}/M$.

2.5 Kähler Structures from Ample Toric Line Bundles

Let X be a smooth projective toric variety over $\mathbb{C}$, and let $L = \mathcal{O}_X(D)$ be an ample T -linearized line bundle on X . Passing to the analytification, we regard L as a holomorphic line bundle over the complex manifold X .

Since L is ample, one can choose a positive smooth Hermitian metric h on L . Let ∇_h be the associated Chern connection, and let $F_h = \nabla_h^2$ be its curvature. With a fixed sign convention, the real $(1, 1)$ -form

$$\omega_D = \sqrt{-1} F_h$$

is a Kähler form on X . In particular, ω_D is a closed non-degenerate real two-form, so (X, ω_D) is a symplectic manifold.

Moreover, since the compact real torus $T_{\mathbb{R}}$ is compact, we may average the Hermitian metric h over $T_{\mathbb{R}}$. Thus h , and hence ω_D , may be chosen to be $T_{\mathbb{R}}$ -invariant. Therefore the algebraic datum $(X, \mathcal{O}_X(D))$ gives rise to a $T_{\mathbb{R}}$ -invariant Kähler, hence symplectic, manifold (X, ω_D) .

Since ω_D is $T_{\mathbb{R}}$ -invariant, the compact torus action on X_{Σ} is symplectic. Using the identifications $\text{Lie}(T_{\mathbb{R}}) \cong N_{\mathbb{R}}$ and $\text{Lie}(T_{\mathbb{R}})^* \cong M_{\mathbb{R}}$, a moment map for this action is a smooth map $\mu_D : X_{\Sigma} \rightarrow M_{\mathbb{R}}$ satisfying

$$d\langle \mu_D, \xi \rangle = \iota_{\xi_X} \omega_D \quad \text{for every } \xi \in N_{\mathbb{R}}.$$

Thus the pair $(X_{\Sigma}, \mathcal{O}_X(D))$ produces a Hamiltonian $T_{\mathbb{R}}$ -space $(X_{\Sigma}, \omega_D, \mu_D)$. The image of μ_D will be discussed in the next section.

2.6 Moment Polytope

Let $X = X_{\Sigma}$ be a smooth projective toric variety, and let $D = \sum_{\rho \in \Sigma(1)} c_{\rho} D_{\rho}$ be a T -invariant Cartier divisor, where D_{ρ} is the torus-invariant divisor corresponding to the ray $\rho = \mathbb{R}_{\geq 0} v_{\rho}$, with $v_{\rho} \in N$ its primitive generator. Assume that $L = \mathcal{O}_X(D)$ is ample.

Definition 2.1 (Moment polytope). The polytope associated to D is

$$\Delta_D = \{m \in M_{\mathbb{R}} \mid \langle m, v_{\rho} \rangle \geq -c_{\rho} \text{ for all } \rho \in \Sigma(1)\}.$$

The main fact we need is the following.

Theorem 2.2. *With the standard normalization determined by D , the image of the moment map $\mu_D : X_{\Sigma} \rightarrow M_{\mathbb{R}}$ associated to the ample line bundle $\mathcal{O}_X(D)$ is precisely the moment polytope:*

$$\mu_D(X_{\Sigma}) = \Delta_D.$$

Proof. We use the projective embedding determined by the complete linear system of $L = \mathcal{O}_X(D)$. By the standard description of global sections of toric line bundles,

$$H^0(X, \mathcal{O}_X(D)) = \bigoplus_{m \in \Delta_D \cap M} \mathbb{C} \cdot \chi^m.$$

Thus the lattice points of Δ_D give a basis of global sections. Writing $\Delta_D \cap M = \{m_0, \dots, m_s\}$, the complete linear system defines a T -equivariant projective embedding $\Phi_D : X_{\Sigma} \rightarrow \mathbb{P}^s$. On the dense torus $T \subset X_{\Sigma}$, this map is given by $t \mapsto [\chi^{m_0}(t) : \dots : \chi^{m_s}(t)]$.

We endow $\mathbb{P}^s$ with the Fubini–Study Kähler form and pull it back to X_{Σ} . The corresponding moment map, restricted to the dense torus, is

$$\mu_D(t) = \frac{\sum_{m \in \Delta_D \cap M} m |\chi^m(t)|^2}{\sum_{m \in \Delta_D \cap M} |\chi^m(t)|^2}.$$

Hence $\mu_D(t)$ is a weighted average of the lattice points $m \in \Delta_D \cap M$, with positive weights $|\chi^m(t)|^2$. It follows that

$$\mu_D(T) \subset \text{Conv}(\Delta_D \cap M) = \Delta_D.$$

Since T is dense in X_{Σ} and μ_D is continuous, we get $\mu_D(X_{\Sigma}) \subset \Delta_D$.

It remains to prove the reverse inclusion. On the dense torus, write $t = \exp(y + i\theta)$, with $y \in N_{\mathbb{R}}$ and $\theta \in N_{\mathbb{R}}/N$. Then $|\chi^m(t)|^2 = e^{2\langle m, y \rangle}$, so the moment map on the open orbit is

$$\mu_D(y) = \frac{\sum_{m \in \Delta_D \cap M} m e^{2\langle m, y \rangle}}{\sum_{m \in \Delta_D \cap M} e^{2\langle m, y \rangle}} = \frac{1}{2} \nabla_y \log \left(\sum_{m \in \Delta_D \cap M} e^{2\langle m, y \rangle} \right).$$

The gradient of the strictly convex function $y \mapsto \log \left(\sum_{m \in \Delta_D \cap M} e^{2\langle m, y \rangle} \right)$ maps $N_{\mathbb{R}}$ onto the interior of the convex hull of the weights $\Delta_D \cap M$, namely onto Δ_D° . Hence $\Delta_D^{\circ} \subset \mu_D(T)$. Taking closures gives $\Delta_D \subset \mu_D(X_{\Sigma})$.

Combining the two inclusions, we obtain

$$\mu_D(X_{\Sigma}) = \Delta_D.$$

□

Thus the ample toric line bundle $\mathcal{O}_X(D)$ determines not only a Kähler form ω_D , but also a canonical convex polytope Δ_D , namely the image of the associated moment map.

3 Categorical and Sheaf-Theoretic Preliminaries

The coherent–constructible correspondence is an equivalence between two derived categories of sheaves. On the coherent side one considers equivariant coherent sheaves on a toric variety, while on the constructible side one considers constructible sheaves whose singular support is constrained by a conic Lagrangian subset determined by the fan. Since morphisms in these categories are naturally derived morphisms, it is not sufficient to regard them merely as ordinary categories. We will therefore use the language of categories enriched over complexes, or equivalently dg categories, in order to keep track of Hom-complexes, shifts, cones, and derived equivalences.

The purpose of this section is only to fix the categorical conventions needed later. We recall the notion of an enriched category, specialize to dg categories, and explain how their homotopy categories recover the triangulated categories appearing in the statement of the CCC. We then review the sheaf-theoretic categories used on both sides of the correspondence: coherent and equivariant coherent sheaves, constructible sheaves, standard and costandard sheaves, singular support, and sheaves with prescribed singular support.

3.1 dg Categories

We first recall the notion of an enriched category. In an ordinary category, the morphisms between two objects form a set. In many geometric situations, however, the morphisms carry additional structure. For example, in derived categories, morphisms are naturally organized into complexes, whose cohomology groups give the corresponding Ext groups. This motivates the notion of a category enriched over a monoidal category.

A **monoidal category** is a category $\mathcal{V}$ together with a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$, a unit object $\mathbf{1} \in \mathcal{V}$, and natural isomorphisms

$$\alpha_{X,Y,Z} : (X \otimes Y) \otimes Z \xrightarrow{\sim} X \otimes (Y \otimes Z), \quad \lambda_X : \mathbf{1} \otimes X \xrightarrow{\sim} X, \quad \rho_X : X \otimes \mathbf{1} \xrightarrow{\sim} X,$$

satisfying the pentagon and triangle axioms. The pentagon axiom is the commutativity of

$$\begin{array}{ccc} ((W \otimes X) \otimes Y) \otimes Z & \xrightarrow{\alpha_{W \otimes X, Y, Z}} & (W \otimes X) \otimes (Y \otimes Z) \xrightarrow{\alpha_{W, X, Y \otimes Z}} W \otimes (X \otimes (Y \otimes Z)) \\ \alpha_{W, X, Y} \otimes \text{id}_Z \downarrow & & \nearrow \text{id}_W \otimes \alpha_{X, Y, Z} \\ (W \otimes (X \otimes Y)) \otimes Z & \xrightarrow{\alpha_{W, X \otimes Y, Z}} & W \otimes ((X \otimes Y) \otimes Z) \end{array}$$

and the triangle axiom is the commutativity of

$$\begin{array}{ccc} (X \otimes \mathbf{1}) \otimes Y & \xrightarrow{\alpha_{X, \mathbf{1}, Y}} & X \otimes (\mathbf{1} \otimes Y) \\ \rho_X \otimes \text{id}_Y \searrow & & \swarrow \text{id}_X \otimes \lambda_Y \\ & X \otimes Y. & \end{array}$$

The bifunctor $\otimes$ is called the **tensor product**.

Basic examples are **Set** with the Cartesian product, **Ab** with the tensor product of abelian groups, and $\text{Ch}(k)$ with the tensor product of complexes of k -modules. In $\text{Ch}(k)$, the unit object is the complex $k[0]$, namely k concentrated in degree zero.

A **$\mathcal{V}$ -enriched category** is a category-like object whose morphisms are objects of $\mathcal{V}$, rather than sets. More precisely, a $\mathcal{V}$ -enriched category $\mathcal{C}$ consists of a collection of objects $\text{Ob}(\mathcal{C})$, an object $\text{Hom}_{\mathcal{C}}(A, B) \in \mathcal{V}$ for every pair $A, B \in \mathcal{C}$, composition morphisms

$$\text{Hom}_{\mathcal{C}}(B, C) \otimes \text{Hom}_{\mathcal{C}}(A, B) \longrightarrow \text{Hom}_{\mathcal{C}}(A, C),$$

and unit morphisms $\mathbf{1} \rightarrow \text{Hom}_{\mathcal{C}}(A, A)$. These data are required to satisfy the usual associativity and unit axioms, written in terms of the monoidal structure of $\mathcal{V}$.

Ordinary categories are recovered by taking $\mathcal{V} = \mathbf{Set}$ with its Cartesian monoidal structure. A **simplicial category** is a category enriched in simplicial sets, so its morphisms are mapping spaces. A **dg category**, short for **differential graded category**, over a field k is a category enriched in $\text{Ch}(k)$, the monoidal category of cochain complexes of k -vector spaces.

For a cochain complex $V^\bullet$, its i -th cohomology is

$$H^i(V^\bullet) = \frac{\ker(d^i : V^i \rightarrow V^{i+1})}{\text{im}(d^{i-1} : V^{i-1} \rightarrow V^i)}.$$

A morphism of complexes $f : V^\bullet \rightarrow W^\bullet$ is called a **quasi-isomorphism** if the induced maps $H^i(f) : H^i(V^\bullet) \rightarrow H^i(W^\bullet)$ are isomorphisms for all $i \in \mathbb{Z}$.

For a dg category $\mathcal{C}$, its **homotopy category** $H^0(\mathcal{C})$ is the ordinary category with the same objects as $\mathcal{C}$, and with morphism spaces

$$\text{Hom}_{H^0(\mathcal{C})}(A, B) = H^0 \text{Hom}_{\mathcal{C}}^\bullet(A, B).$$

A **dg functor** $F : \mathcal{C} \rightarrow \mathcal{D}$ sends objects to objects and induces morphisms of complexes

$$\mathrm{Hom}_{\mathcal{C}}^{\bullet}(A, B) \longrightarrow \mathrm{Hom}_{\mathcal{D}}^{\bullet}(F(A), F(B)),$$

compatible with composition and units.

A dg functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called a **quasi-equivalence** if, for all objects $A, B \in \mathcal{C}$, the induced morphism of Hom-complexes is a quasi-isomorphism, and if the induced functor $H^0(F) : H^0(\mathcal{C}) \rightarrow H^0(\mathcal{D})$ is essentially surjective. Equivalently, for every object $D \in \mathcal{D}$, there exists an object $C \in \mathcal{C}$ such that $F(C) \cong D$ in $H^0(\mathcal{D})$.

A **right dg module** over $\mathcal{C}$ is a dg functor $M : \mathcal{C}^{op} \rightarrow \mathrm{Ch}(k)$. Equivalently, it assigns to each object $A \in \mathcal{C}$ a complex $M(A)$, together with action maps

$$M(B) \otimes \mathrm{Hom}_{\mathcal{C}}^{\bullet}(A, B) \longrightarrow M(A),$$

compatible with composition and units. We denote by $\mathrm{Mod}_{dg}(\mathcal{C})$ the dg category of right dg modules over $\mathcal{C}$.

For each object $A \in \mathcal{C}$, the associated **right representable dg module** is $h_A := \mathrm{Hom}_{\mathcal{C}}^{\bullet}(-, A)$. Explicitly, $h_A(X) = \mathrm{Hom}_{\mathcal{C}}^{\bullet}(X, A)$, and a morphism $u : X \rightarrow Y$ acts by precomposition:

$$\mathrm{Hom}_{\mathcal{C}}^{\bullet}(Y, A) \longrightarrow \mathrm{Hom}_{\mathcal{C}}^{\bullet}(X, A), \quad g \longmapsto g \circ u.$$

If $V^{\bullet}$ is a cochain complex, its **shift** $V[1]$ is defined by $V[1]^i = V^{i+1}$ and $d_{V[1]}^i = -d_V^{i+1}$. If $f : V^{\bullet} \rightarrow W^{\bullet}$ is a morphism of complexes, its **cone** is the complex

$$\mathrm{Cone}(f)^i = W^i \oplus V^{i+1}, \quad d(w, v) = (d_W w + f(v), -d_V v).$$

The dg category $\mathrm{Mod}_{dg}(\mathcal{C})$ has shifts and cones defined objectwise. For example, $M[1](A) = M(A)[1]$, and if $\varphi : M \rightarrow N$ is a closed degree-zero morphism of dg modules, then

$$\mathrm{Cone}(\varphi)(A) = \mathrm{Cone}(M(A) \rightarrow N(A)).$$

A dg category $\mathcal{C}$ is called **pretriangulated** if shifts and cones of representable dg modules are again representable. More precisely, for every $A \in \mathcal{C}$, the shifted module $h_A[1]$ is represented by an object of $\mathcal{C}$, and for every closed degree-zero morphism $f : A \rightarrow B$, the cone of the induced morphism $h_A \rightarrow h_B$ is also represented by an object of $\mathcal{C}$. In this case, $H^0(\mathcal{C})$ becomes a triangulated category.

Not every dg category is pretriangulated. Let $\mathcal{C}$ be the dg category with one object $*$ and endomorphism complex $\mathrm{Hom}_{\mathcal{C}}(*, *) = k$, where k is concentrated in degree zero. The unique representable right dg module is $h_* = \mathrm{Hom}_{\mathcal{C}}(-, *) \simeq k$. However, the shifted module $h_*[1] \simeq k[1]$ is not representable by an object of $\mathcal{C}$, since $\mathcal{C}$ has only the object $*$, whose representable module is k , not $k[1]$. Therefore $\mathcal{C}$ is not pretriangulated.

The **dg Yoneda embedding** is the dg functor

$$Y : \mathcal{C} \longrightarrow \mathrm{Mod}_{dg}(\mathcal{C}), \quad A \longmapsto \mathrm{Hom}_{\mathcal{C}}^{\bullet}(-, A).$$

The **pretriangulated envelope** of $\mathcal{C}$, denoted $\mathrm{Pretr}(\mathcal{C})$, is the smallest full dg subcategory of $\mathrm{Mod}_{dg}(\mathcal{C})$ which contains all representable dg modules h_A and is closed under finite direct sums, shifts, and cones.

For the one-object dg category above, the pretriangulated envelope is obtained by adjoining formal shifts, cones, and finite direct sums. After idempotent completion, it is identified with $\mathrm{Perf}(k)$, the dg category of perfect complexes over k .

Let $\mathcal{C}$ be a pretriangulated dg category. A full dg subcategory $\mathcal{A} \subset \mathcal{C}$ is called **thick** if $H^0(\mathcal{A})$ is a full triangulated subcategory of $H^0(\mathcal{C})$ closed under direct summands. Equivalently, if $X \in H^0(\mathcal{A})$ and Y is a retract of X in $H^0(\mathcal{C})$, then $Y \in H^0(\mathcal{A})$.

For a collection of objects $\mathcal{S} \subset \mathrm{Ob}(\mathcal{C})$, we denote by $\langle \mathcal{S} \rangle$ the smallest full dg subcategory of $\mathcal{C}$ whose homotopy category is a thick triangulated subcategory of $H^0(\mathcal{C})$ containing $\mathcal{S}$.

Let $\mathcal{C}$ and $\mathcal{D}$ be pretriangulated dg categories. A dg functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called **exact** if the induced functor $H^0(F) : H^0(\mathcal{C}) \rightarrow H^0(\mathcal{D})$ is an exact functor of triangulated categories. Equivalently, it commutes with shifts up to isomorphism and sends distinguished triangles to distinguished triangles.

In practice, we will use the following criterion.

Theorem 3.1 (Generator Criterion for Quasi-Equivalences). *Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an exact dg functor between pretriangulated dg categories. Let $\mathcal{S} \subset \mathrm{Ob}(\mathcal{C})$ be a collection of objects such that $\langle \mathcal{S} \rangle = \mathcal{C}$ and $\langle F(\mathcal{S}) \rangle = \mathcal{D}$. Suppose that, for all $A, B \in \mathcal{S}$, the induced morphism of complexes*

$$\mathrm{Hom}_{\mathcal{C}}^{\bullet}(A, B) \longrightarrow \mathrm{Hom}_{\mathcal{D}}^{\bullet}(F(A), F(B))$$

is a quasi-isomorphism. Then F is a quasi-equivalence.

Therefore, the proof of a dg quasi-equivalence can often be reduced to two tasks:

1. compare the Hom-complexes on a set of generators;
2. prove that these generators generate both sides.

3.2 Coherent and Equivariant Coherent Sheaves

Let X be a locally Noetherian scheme. We denote by $\text{Coh}(X)$ the abelian category of coherent $\mathcal{O}_X$ -modules. Let $\text{Ch}^b(\text{Coh}(X))$ be the category of bounded complexes of coherent sheaves, and let $K^b(\text{Coh}(X))$ be its homotopy category. Thus $K^b(\text{Coh}(X))$ has the same objects as $\text{Ch}^b(\text{Coh}(X))$, but its morphisms are chain maps modulo chain homotopy.

The **bounded derived category** of coherent sheaves is the Verdier localization

$$D^b(\text{Coh}(X)) = K^b(\text{Coh}(X))[\mathcal{W}^{-1}],$$

where $\mathcal{W}$ denotes the class of quasi-isomorphisms in $K^b(\text{Coh}(X))$.

Now suppose that an algebraic group G acts on X . Let $a : G \times X \rightarrow X$ be the action map, $p_X : G \times X \rightarrow X$ the projection to X , and $\mu : G \times G \rightarrow G$ the multiplication map. On $G \times G \times X$, define

$$q_X(g, h, x) = x, \quad q_{23}(g, h, x) = (h, x), \quad (\text{id}_G \times a)(g, h, x) = (g, hx),$$

and regard $\mu \times \text{id}_X : G \times G \times X \rightarrow G \times X$ as the map $(g, h, x) \mapsto (gh, x)$.

A **G -equivariant coherent sheaf** on X is a coherent sheaf $\mathcal{F}$ together with an isomorphism

$$\phi : p_X^* \mathcal{F} \xrightarrow{\sim} a^* \mathcal{F}$$

satisfying the following cocycle condition:

$$\begin{array}{ccc} q_X^* \mathcal{F} & \xrightarrow{q_{23}^* \phi} & (a \circ q_{23})^* \mathcal{F} \\ & \searrow (\mu \times \text{id}_X)^* \phi & \downarrow (\text{id}_G \times a)^* \phi \\ & & (a \circ (\text{id}_G \times a))^* \mathcal{F}. \end{array}$$

Here we use the equality $a \circ (\text{id}_G \times a) = a \circ (\mu \times \text{id}_X)$ as maps $G \times G \times X \rightarrow X$. We call such an isomorphism ϕ a **G -linearization of $\mathcal{F}$** .

We denote by $\text{Coh}_G(X)$ the **abelian category of G -equivariant coherent sheaves on X** , and by $D^b(\text{Coh}_G(X))$ its **bounded equivariant derived category**.

A complex $\mathcal{E}^\bullet$ of $\mathcal{O}_X$ -modules is called **perfect** if it is locally quasi-isomorphic to a bounded complex of finite rank locally free sheaves. We write $\text{Perf}(X)$ for the **dg category of perfect complexes on X** . If G acts on X , we similarly write $\text{Perf}_G(X)$ for the **dg category of G -equivariant perfect complexes**.

The following standard fact explains why perfect complexes provide the correct dg model for the coherent side of the correspondence. In the smooth case, bounded coherent complexes can be represented by perfect complexes. In the equivariant setting, the same statement holds provided that there are sufficiently many equivariant vector bundles.

Theorem 3.2. *Let X be a quasi-compact and quasi-separated scheme. Then*

$$D(\text{QCoh}(X))^c \simeq \text{Perf}(X).$$

Equivariantly, for the toric varieties considered in this note, one has

$$D(\text{QCoh}_T(X_\Sigma))^c \simeq \text{Perf}_T(X_\Sigma).$$

Theorem 3.3. *Let X be a smooth Noetherian scheme over a field k .*

1. *The dg category $\text{Perf}(X)$ is a dg enhancement of $D^b(\text{Coh}(X))$. Equivalently,*

$$H^0(\text{Perf}(X)) \simeq D^b(\text{Coh}(X)).$$

2. *Let G be an algebraic group acting on X . If X has the equivariant resolution property, then $\text{Perf}_G(X)$ is a dg enhancement of $D^b(\text{Coh}_G(X))$. Equivalently,*

$$H^0(\text{Perf}_G(X)) \simeq D^b(\text{Coh}_G(X)).$$

Here, we say that X has the **equivariant resolution property** if every G -equivariant coherent sheaf $\mathcal{F} \in \text{Coh}_G(X)$ admits a surjection

$$\mathcal{E} \twoheadrightarrow \mathcal{F}$$

from a G -equivariant vector bundle $\mathcal{E}$ on X .

3.3 Constructible Sheaves

Let M be a real analytic manifold, and let k be a fixed field of coefficients. We denote by $Sh(M)$ the **dg category of complexes of sheaves of k -vector spaces on M** . Equivalently, $Sh(M)$ may be regarded as a dg enhancement of the derived category of sheaves of k -vector spaces on M .

We first recall the notion of a locally constant sheaf. A sheaf $\mathcal{F}$ of k -vector spaces on M is called **locally constant** if every point $x \in M$ has an open neighborhood U such that $\mathcal{F}|_U \simeq k_U^{\oplus r}$ for some integer r depending on U . A **local system** is a locally constant sheaf whose stalks are finite-dimensional k -vector spaces.

A **stratification** of M is a locally finite decomposition $M = \bigsqcup_{\alpha \in A} S_\alpha$ into locally closed smooth submanifolds. A sheaf $\mathcal{F}$ is **constructible** with respect to this stratification if each restriction $\mathcal{F}|_{S_\alpha}$ is a local system. A sheaf is called constructible if it is constructible with respect to some such stratification.

We denote by $Sh_c(M)$ the full dg subcategory of $Sh(M)$ consisting of constructible complexes. More precisely, a complex $\mathcal{F}^\bullet \in Sh(M)$ is constructible if each cohomology sheaf $\mathcal{H}^i(\mathcal{F}^\bullet)$ is constructible and $\mathcal{H}^i(\mathcal{F}^\bullet) = 0$ for all but finitely many i . We write $Sh_{cc}(M)$ for the full dg subcategory of $Sh_c(M)$ consisting of constructible complexes with compact support. Here the **support of a complex $\mathcal{F}^\bullet$** is the union of the supports of its cohomology sheaves.

For a fixed stratification $\mathcal{S} = \{S_\alpha\}_{\alpha \in A}$, we write $D_S^b(M; k)$ for the bounded derived category of k -sheaves constructible with respect to $\mathcal{S}$.

Let $j : U \hookrightarrow M$ be the inclusion of a locally closed subset. We denote by k_U the constant sheaf on U with coefficients in k . The **standard sheaf associated to U** is $j_! k_U$, where $j_!$ denotes extension by zero. The **costandard sheaf associated to U** is $j_* k_U$, where j_* denotes the usual direct image.

Standard and costandard sheaves are the basic building blocks of constructible categories. The following generation statement is the constructible-side analogue of choosing generators on the coherent side.

Theorem 3.4 (Generation by standard and costandard sheaves). *Let M be a real analytic manifold, and let $M = \bigsqcup_{\alpha \in A} S_\alpha$ be a finite stratification by locally closed submanifolds. Let $j_\alpha : S_\alpha \hookrightarrow M$ denote the inclusion. Then*

$$D_S^b(M; k) = \langle j_{\alpha!} L \mid \alpha \in A, L \text{ a finite-dimensional local system on } S_\alpha \rangle.$$

Equivalently,

$$D_S^b(M; k) = \langle j_{\alpha*} L \mid \alpha \in A, L \text{ a finite-dimensional local system on } S_\alpha \rangle.$$

In particular, if every stratum S_α is contractible, then

$$D_S^b(M; k) = \langle j_{\alpha!} k_{S_\alpha} \mid \alpha \in A \rangle = \langle j_{\alpha*} k_{S_\alpha} \mid \alpha \in A \rangle.$$

3.4 Singular Support

Let M be a real analytic manifold, and let T^*M denote its cotangent bundle. The cotangent bundle carries a natural fiberwise dilation action of $\mathbb{R}_{>0}$, given by $t \cdot (x, \xi) = (x, t\xi)$. A subset $\Lambda \subset T^*M$ is called **conic** if it is invariant under this action; equivalently, $(x, \xi) \in \Lambda$ implies $(x, t\xi) \in \Lambda$ for every $t \in \mathbb{R}_{>0}$.

For a constructible complex of sheaves $\mathcal{F} \in Sh_c(M)$, its **singular support**, also called its **microsupport**, is a closed conic subset $SS(\mathcal{F}) \subset T^*M$. It is a microlocal refinement of the ordinary support. A covector $(x_0, \xi_0) \in T^*M$ is not contained in $SS(\mathcal{F})$ if there exists a conic open neighborhood $U \subset T^*M$ of (x_0, ξ_0) such that, for every C^1 -function φ defined near x_0 with $\varphi(x_0) = 0$ and $(x_0, d\varphi_{x_0}) \in U$, one has

$$(R\Gamma_{\{\varphi \geq 0\}}(\mathcal{F}))_{x_0} = 0.$$

Here $R\Gamma_Z(\mathcal{F})$ denotes the derived functor of sections with support in a closed subset $Z \subset M$. Thus $(R\Gamma_{\{\varphi \geq 0\}}(\mathcal{F}))_{x_0}$ is the local cohomology complex of $\mathcal{F}$ at x_0 with support in the half-space $\{\varphi \geq 0\}$.

Equivalently, $SS(\mathcal{F})$ is the closed conic subset of T^*M consisting of covectors along which this local cohomological vanishing fails. It records the cotangent directions in which sections of $\mathcal{F}$ cannot be propagated microlocally.

If $\mathcal{F}$ is locally constant on M , then it has no nonzero singular directions; hence $SS(\mathcal{F}) \subset T_M^*M$, where $T_M^*M \subset T^*M$ denotes the zero section.

Let $S \subset M$ be a locally closed submanifold. The **conormal bundle** of S in M is

$$T_S^*M = \{(x, \xi) \in T^*M \mid x \in S, \xi|_{T_x S} = 0\}.$$

If $\mathcal{F}$ is constructible with respect to a stratification $M = \bigsqcup_{\alpha \in A} S_\alpha$, then

$$SS(\mathcal{F}) \subset \bigcup_{\alpha \in A} T_{S_\alpha}^*M.$$

This gives the basic geometric control on the singularities of a constructible sheaf.

Let $\Lambda \subset T^*M$ be a closed conic subset. We define

$$Sh_\Lambda(M) := \{\mathcal{F} \in Sh_c(M) \mid SS(\mathcal{F}) \subset \Lambda\}$$

to be the full dg subcategory of $Sh_c(M)$ consisting of constructible complexes whose singular support is contained in Λ . Thus Λ imposes a microlocal constraint on constructible sheaves: instead of allowing arbitrary constructible singularities, one only allows singular behavior in the cotangent directions prescribed by Λ .

Theorem 3.5. *Let $\Lambda \subset T^*M$ be a closed conic subset. The full subcategory $Sh_\Lambda(M) \subset Sh_c(M)$ is closed under shifts, cones, finite direct sums, and retracts. Consequently, $H^0(Sh_\Lambda(M))$ is a triangulated subcategory of $H^0(Sh_c(M))$.*

Theorem 3.6 (Microlocal Morse lemma). *Let M be a real analytic manifold, let $F \in Sh_c(M)$, and let $f : M \rightarrow \mathbb{R}$ be a real analytic function. Let $a < b$. Suppose that f is proper on $\text{supp}(F) \cap f^{-1}([a, b])$, and that*

$$df(x) \notin SS(F)$$

for every $x \in \text{supp}(F) \cap f^{-1}([a, b])$. Then the restriction map

$$R\Gamma(f^{-1}((-\infty, b)); F) \longrightarrow R\Gamma(f^{-1}((-\infty, a)); F)$$

is a quasi-isomorphism.

Thus singular support controls the cotangent directions in which local cohomology may fail to propagate. In directions outside $SS(F)$, passing through nearby level sets of f does not create new cohomology.

4 The Coherent Side for Toric Varieties

In this section we specialize the coherent side of the coherent-constructible correspondence to the toric setting. Let X_Σ be a smooth toric variety with acting torus T , and let M and N denote the character and cocharacter lattices of T . The coherent side of the correspondence is modeled by the dg category

$$\text{Perf}_T(X_\Sigma),$$

or equivalently, in the smooth setting, by the bounded equivariant coherent derived category $D^b(\text{Coh}_T(X_\Sigma))$.

The toric structure gives a particularly explicit description of this category. On an affine toric chart $U_\sigma = \text{Spec } k[\sigma^\vee \cap M]$, T -equivariant quasi-coherent sheaves correspond to M -graded modules over the semigroup algebra $k[\sigma^\vee \cap M]$. Thus equivariance translates the torus action into a grading by the character lattice M .

The purpose of this section is to recall this graded description and to introduce the coherent generators used in the proof of the toric CCC. These generators are indexed by pairs (σ, χ) , where $\sigma \in \Sigma$ is a cone of the fan and $\chi \in M$ is a character. They will later be matched with standard constructible sheaves on $M_\mathbb{R}$. The main result prepared in this section is that these coherent objects generate $\text{Perf}_T(X_\Sigma)$.

4.1 Equivariant Coherent Sheaves on Toric Varieties

Let $\sigma \in \Sigma$ be a cone, and let $U_\sigma \subset X_\Sigma$ be the corresponding affine toric open subset. We write $U_\sigma = \text{Spec } A_\sigma$, where $A_\sigma = k[\sigma^\vee \cap M]$. The semigroup algebra A_σ has its natural M -grading

$$A_\sigma = \bigoplus_{m \in \sigma^\vee \cap M} A_{\sigma, m}, \quad A_{\sigma, m} = k \cdot \chi^m.$$

Equivalently, $\deg(\chi^m) = m$. This grading records the torus action on U_σ .

We denote by $A_\sigma\text{-Mod}^M$ the category of M -graded A_σ -modules. Thus an object is an A_σ -module $P = \bigoplus_{\chi \in M} P_\chi$ such that $A_{\sigma, m} \cdot P_\chi \subset P_{\chi+m}$ for all $m \in \sigma^\vee \cap M$ and $\chi \in M$.

Proposition 4.1. *There is an equivalence of categories*

$$\text{QCoh}_T(U_\sigma) \simeq A_\sigma\text{-Mod}^M.$$

Proof. Since U_σ is affine, a quasi-coherent sheaf is determined by its module of global sections. If $\mathcal{F}$ is T -equivariant, then $\Gamma(U_\sigma, \mathcal{F})$ carries a compatible T -action. Since T is a torus, this module decomposes into weight spaces:

$$\Gamma(U_\sigma, \mathcal{F}) = \bigoplus_{\chi \in M} \Gamma(U_\sigma, \mathcal{F})_\chi.$$

Compatibility with the A_σ -module structure is precisely the grading condition

$$A_{\sigma, m} \cdot \Gamma(U_\sigma, \mathcal{F})_\chi \subset \Gamma(U_\sigma, \mathcal{F})_{\chi+m}.$$

Thus $\Gamma(U_\sigma, \mathcal{F})$ is an M -graded A_σ -module.

Conversely, an M -graded A_σ -module P defines a quasi-coherent sheaf $\tilde{P}$ on U_σ , and the grading defines a T -equivariant structure on $\tilde{P}$. These two constructions are inverse to each other. $\square$

Let $A_\sigma\text{-mod}^M$ denote the full subcategory of finitely generated M -graded A_σ -modules.

Corollary 4.2. *There is an equivalence of categories*

$$\text{Coh}_T(U_\sigma) \simeq A_\sigma\text{-mod}^M.$$

Proof. By Proposition 4.1, T -equivariant quasi-coherent sheaves on U_σ are equivalent to M -graded A_σ -modules. Since U_σ is affine, coherence is equivalent to finite generation over A_σ . Hence the coherent objects correspond exactly to finitely generated graded modules. $\square$

We now pass from a single affine chart to the whole toric variety. The fan Σ gives an affine open cover $X_\Sigma = \bigcup_{\sigma \in \Sigma} U_\sigma$, and for two cones $\sigma, \tau \in \Sigma$, one has $U_\sigma \cap U_\tau = U_{\sigma \cap \tau}$. The open immersions $U_{\sigma \cap \tau} \hookrightarrow U_\sigma$ and $U_{\sigma \cap \tau} \hookrightarrow U_\tau$ induce morphisms of M -graded algebras $A_\sigma \rightarrow A_{\sigma \cap \tau}$ and $A_\tau \rightarrow A_{\sigma \cap \tau}$.

Proposition 4.3 (Gluing over the fan). *A T -equivariant quasi-coherent sheaf on X_Σ is equivalently given by a family of M -graded A_σ -modules P_σ , one for each cone $\sigma \in \Sigma$, together with isomorphisms of M -graded $A_{\sigma \cap \tau}$ -modules*

$$A_{\sigma \cap \tau} \otimes_{A_\sigma} P_\sigma \xrightarrow{\sim} A_{\sigma \cap \tau} \otimes_{A_\tau} P_\tau$$

for all pairs $\sigma, \tau \in \Sigma$, satisfying the usual cocycle condition on triple overlaps. The corresponding sheaf is coherent if and only if each P_σ is a finitely generated M -graded A_σ -module.

Proof. This is the usual descent description of quasi-coherent sheaves with respect to the affine open cover $\{U_\sigma\}_{\sigma \in \Sigma}$, combined with the equivalence $\mathrm{QCoh}_T(U_\sigma) \simeq A_\sigma\text{-Mod}^M$ on each affine chart. Restricting a sheaf from U_σ to $U_{\sigma \cap \tau}$ corresponds algebraically to extension of scalars along $A_\sigma \rightarrow A_{\sigma \cap \tau}$. Since these restriction maps preserve the M -grading, the gluing maps are isomorphisms of graded modules. The cocycle condition is exactly the usual sheaf gluing condition on triple overlaps. Finally, coherence is local on X_Σ , so the coherent case is obtained by requiring each P_σ to be finitely generated. $\square$

Thus T -equivariant coherent sheaves on X_Σ can be described in terms of M -graded modules over the affine semigroup algebras A_σ , together with compatibility over the fan. This local-to-global description will be used in the construction of the coherent generators.

4.2 Coherent Generators

We now introduce the coherent generators used in the toric coherent-constructible correspondence. For each cone $\sigma \in \Sigma$, let $i_\sigma : U_\sigma \hookrightarrow X_\Sigma$ be the inclusion of the corresponding affine toric open subset.

Definition 4.4. Let $\sigma \in \Sigma$, and let U_σ be the corresponding affine toric open subset. For a character $\chi \in M$, define $\mathcal{O}_{U_\sigma}(\chi)$ to be the trivial line bundle $U_\sigma \times k \rightarrow U_\sigma$ with the following T -action on its total space:

$$t \cdot (x, v) = (tx, \chi(t)v), \quad t \in T, \quad x \in U_\sigma, \quad v \in k.$$

Proposition 4.5. *The line bundle $\mathcal{O}_{U_\sigma}(\chi)$ is a T -equivariant line bundle on U_σ .*

Proof. Since $\chi : T \rightarrow k^*$ is a character, it is a group homomorphism. Hence, for $t_1, t_2 \in T$, one has

$$t_1 \cdot (t_2 \cdot (x, v)) = t_1 \cdot (t_2 x, \chi(t_2)v) = (t_1 t_2 x, \chi(t_1)\chi(t_2)v) = (t_1 t_2 x, \chi(t_1 t_2)v).$$

Thus the formula defines a T -action on $U_\sigma \times k$.

Let $\pi : U_\sigma \times k \rightarrow U_\sigma$ be the projection. Then

$$\pi(t \cdot (x, v)) = \pi(tx, \chi(t)v) = tx = t \cdot \pi(x, v),$$

so π is T -equivariant. Moreover, for each $x \in U_\sigma$, the induced map on fibers

$$(\mathcal{O}_{U_\sigma}(\chi))_x \longrightarrow (\mathcal{O}_{U_\sigma}(\chi))_{tx}$$

is multiplication by $\chi(t)$, hence is k -linear. Therefore $\mathcal{O}_{U_\sigma}(\chi)$ is a T -equivariant line bundle. $\square$

We call this equivariant structure the **T -linearization of weight χ** . Thus $\mathcal{O}_{U_\sigma}(\chi)$ has the same underlying line bundle as $\mathcal{O}_{U_\sigma}$, but its T -equivariant structure is twisted by the character χ .

Remark 4.6. The word “linearization” means that the T -action on U_σ is lifted to the total space of the line bundle, with linear maps on fibers. Here the induced map

$$(\mathcal{O}_{U_\sigma})_x \longrightarrow (\mathcal{O}_{U_\sigma})_{tx}$$

is multiplication by $\chi(t)$. Equivalently, this is descent data for the action groupoid $T \times U_\sigma \rightrightarrows U_\sigma$.

We define

$$\Theta'(\sigma, \chi) := (i_\sigma)_* \mathcal{O}_{U_\sigma}(\chi).$$

These objects are T -equivariant quasi-coherent sheaves on X_Σ . They are indexed by the same combinatorial data (σ, χ) which will later index the constructible generators.

Theorem 4.7 (Theta generation in the equivariant quasi-coherent category). *Let X_Σ be a smooth toric variety. Then*

$$D(\mathrm{QCoh}_T(X_\Sigma)) = \left\langle \Theta'(\sigma, \chi) \mid \sigma \in \Sigma, \chi \in M \right\rangle_{\mathrm{loc}}.$$

Here $\langle - \rangle_{\mathrm{loc}}$ denotes the smallest localizing triangulated subcategory containing the indicated objects.

Proof. Let $\mathcal{C}$ be the right-hand side. We prove that every object of $D(\mathrm{QCoh}_T(X_\Sigma))$ belongs to $\mathcal{C}$.

First consider an affine toric chart $U_\sigma = \mathrm{Spec} A_\sigma$, where $A_\sigma = k[\sigma^\vee \cap M]$. By [Proposition 4.1](#), one has $\mathrm{QCoh}_T(U_\sigma) \simeq A_\sigma\text{-Mod}^M$. Under this equivalence, $\mathcal{O}_{U_\sigma}(\chi)$ corresponds to the free graded module $A_\sigma(\chi)$. The modules $A_\sigma(\chi)$, for $\chi \in M$, generate $D(A_\sigma\text{-Mod}^M)$: if $P \in D(A_\sigma\text{-Mod}^M)$ satisfies $\mathrm{RHom}_{A_\sigma}(A_\sigma(\chi), P) = 0$ for every $\chi \in M$, then all graded components of all cohomology modules $H^i(P)$ vanish, hence $P \simeq 0$.

Now let $F \in D(\mathrm{QCoh}_T(X_\Sigma))$. Choose the finite T -stable affine cover of X_Σ by affine toric charts. Since intersections of affine toric charts are again affine toric charts, the associated Čech resolution expresses F as a finite totalization of objects of the form $(i_\tau)_*(F|_{U_\tau})$, with $\tau \in \Sigma$. For each τ , the restriction $F|_{U_\tau}$ is generated by $\mathcal{O}_{U_\tau}(\chi)$, $\chi \in M$. Since $(i_\tau)_*$ is exact and preserves coproducts for these quasi-compact open immersions, $(i_\tau)_*(F|_{U_\tau})$ is generated by $(i_\tau)_*\mathcal{O}_{U_\tau}(\chi) = \Theta'(\tau, \chi)$. Hence every term in the Čech resolution belongs to $\mathcal{C}$.

Since $\mathcal{C}$ is localizing, it is closed under shifts, cones, finite totalizations, and coproducts. Therefore $F \in \mathcal{C}$. This proves

$$D(\mathrm{QCoh}_T(X_\Sigma)) = \left\langle \Theta'(\sigma, \chi) \mid \sigma \in \Sigma, \chi \in M \right\rangle_{\mathrm{loc}}.$$

□

Corollary 4.8 (Compact form of the coherent side). *Let X_Σ be a smooth toric variety. Then the compact part of the equivariant quasi-coherent derived category is the perfect equivariant category:*

$$D(\mathrm{QCoh}_T(X_\Sigma))^c \simeq \mathrm{Perf}_T(X_\Sigma).$$

Proof. This follows directly from [Theorem 4.7](#) and [Theorem 3.2](#). □

Thus the equivariant quasi-coherent side has a distinguished family of theta generators indexed by pairs (σ, χ) . The perfect coherent side will be obtained by passing to compact objects as in [Corollary 4.8](#).

5 The Constructible Side and the FLTZ Skeleton

Throughout this section, let Σ be a rational polyhedral fan in $N_\mathbb{R}$. No smoothness assumption is imposed here, since the constructions in this section depend only on the polyhedral data of Σ . The constructible side is not formulated on the toric variety X_Σ , but on the real vector space $M_\mathbb{R}$. Under the natural identification

$$T^*M_\mathbb{R} \cong M_\mathbb{R} \times (M_\mathbb{R})^* \cong M_\mathbb{R} \times N_\mathbb{R},$$

the fan $\Sigma \subset N_\mathbb{R}$ appears as a source of prescribed cotangent directions. Thus this section constructs the microlocal object that controls the constructible side of the correspondence.

5.1 The FLTZ Skeleton and Prescribed Singular Support

We first define the FLTZ skeleton associated to Σ . In the above identification of the cotangent bundle, the first factor is the base space and the second factor records cotangent directions. Thus the fan in $N_\mathbb{R}$ imposes a microlocal condition on sheaves on $M_\mathbb{R}$.

Definition 5.1 (FLTZ skeleton). For a cone $\sigma \in \Sigma$, set

$$\sigma^\perp = \{m \in M_\mathbb{R} \mid \langle m, n \rangle = 0 \text{ for all } n \in \sigma\}, \quad M_\sigma = M/(\sigma^\perp \cap M).$$

For $\chi \in M_\sigma$, let $\chi + \sigma^\perp$ denote the affine subspace of $M_\mathbb{R}$ determined by any lift of χ to M . This is well-defined because two lifts differ by an element of $\sigma^\perp \cap M$.

The FLTZ skeleton associated to Σ is the subset

$$\Lambda_\Sigma = \bigcup_{\sigma \in \Sigma} \bigcup_{\chi \in M_\sigma} (\chi + \sigma^\perp) \times (-\sigma) \subset T^*M_\mathbb{R}.$$

Equivalently,

$$\Lambda_\Sigma = \bigcup_{\sigma \in \Sigma} (\sigma^\perp + M) \times (-\sigma).$$

Here $\chi + \sigma^\perp$ records the base locus in $M_\mathbb{R}$, while $-\sigma$ records the allowed cotangent directions.

Proposition 5.2. *The subset Λ_Σ is conic with respect to the cotangent fibers.*

Proof. The conic action on $T^*M_{\mathbb{R}} \cong M_{\mathbb{R}} \times N_{\mathbb{R}}$ scales only the second factor. If

$$(m, n) \in (\chi + \sigma^\perp) \times (-\sigma),$$

then for every $t \in \mathbb{R}_{\geq 0}$, the base point m is unchanged and $tn \in -\sigma$, since $-\sigma$ is a cone. Hence

$$(m, tn) \in (\chi + \sigma^\perp) \times (-\sigma) \subset \Lambda_\Sigma.$$

Therefore Λ_Σ is conic in the cotangent direction. $\square$

Using the notion of singular support recalled in Section 3, define

$$Sh_{\Lambda_\Sigma}(M_{\mathbb{R}}) = \{F \in Sh(M_{\mathbb{R}}) \mid SS(F) \subset \Lambda_\Sigma\}.$$

This is the category of constructible sheaves on $M_{\mathbb{R}}$ whose singular support is prescribed by the FLTZ skeleton.

5.2 Singular Support of Polyhedral Sheaves

We first record the singular-support estimate for costandard sheaves on closed convex polyhedra.

Lemma 5.3 (Singular support of a polyhedral costandard sheaf). *Let $P \subset M_{\mathbb{R}}$ be a closed convex polyhedron, and let $i_P : P \hookrightarrow M_{\mathbb{R}}$ be the inclusion. Put*

$$\omega_P := i_P^! k_{M_{\mathbb{R}}}.$$

For a face $G \preceq P$, define the normal cone

$$N_GP := \{n \in N_{\mathbb{R}} \mid \langle m - g, n \rangle \geq 0 \text{ for all } m \in P, g \in G\}.$$

Then

$$SS((i_P)_* \omega_P) \subset \bigcup_{G \preceq P} G \times (-N_GP) \subset T^*M_{\mathbb{R}}.$$

Proof. First consider k_P . Stratify P by its relative open faces. Near a point of $\text{relint}(G)$, the pair $(M_{\mathbb{R}}, P)$ is microlocally modeled by the tangent cone to P along G . The local computation for a closed convex polyhedral cone says that the singular covectors of the constant sheaf on this cone are contained in its polar cone. Hence, along the face G , the singular covectors of k_P are contained in N_GP . Therefore

$$SS(k_P) \subset \bigcup_{G \preceq P} G \times N_GP.$$

By Verdier duality,

$$(i_P)_* \omega_P \simeq \mathbb{D}(k_P).$$

Moreover,

$$SS(\mathbb{D}F) = SS(F)^a, \quad (x, \xi)^a = (x, -\xi).$$

Thus

$$SS((i_P)_* \omega_P) = SS(\mathbb{D}k_P) \subset \bigcup_{G \preceq P} G \times (-N_GP).$$

$\square$

Proposition 5.4. *Let $\sigma \in \Sigma$, $\chi \in M_\sigma$, and set $P_{\sigma, \chi} := \chi + \sigma^\vee$. Then*

$$SS((i_{\sigma, \chi})_* \omega_{P_{\sigma, \chi}}) \subset \bigcup_{\tau \preceq \sigma} (\chi + \tau^\perp) \times (-\tau) \subset \Lambda_\Sigma.$$

Proof. The faces of $P_{\sigma, \chi}$ are indexed by faces $\tau \preceq \sigma$. The face corresponding to τ is contained in $\chi + \tau^\perp$, and its normal cone is τ . Applying the polyhedral singular-support estimate gives

$$SS((i_{\sigma, \chi})_* \omega_{P_{\sigma, \chi}}) \subset \bigcup_{\tau \preceq \sigma} (\chi + \tau^\perp) \times (-\tau).$$

Since every face $\tau \preceq \sigma$ is again a cone of Σ , the right-hand side is contained in Λ_Σ . $\square$

Lemma 5.5 (Polyhedral microlocal dévissage). *Let $\mathcal{C}_\Sigma$ be the stable idempotent-complete subcategory of $Sh_{\Lambda_\Sigma}(M_{\mathbb{R}})$ generated by the costandard sheaves*

$$(i_{\sigma, \chi})_* \omega_{\chi + \sigma^\vee}, \quad \sigma \in \Sigma, \chi \in M_\sigma.$$

Then every object of $Sh_{\Lambda_\Sigma}(M_{\mathbb{R}})$ belongs to $\mathcal{C}_\Sigma$.

Sketch of proof. The condition $SS(F) \subset \Lambda_\Sigma$ implies that the singularities of F can occur only along affine subspaces of the form $\chi + \sigma^\perp$, with conormal directions contained in $-\sigma$. Choose a polyhedral stratification of $M_{\mathbb{R}}$ compatible with these affine subspaces. By the constructible dévissage theorem, F is generated by standard or costandard sheaves on the strata. The microlocal condition excludes all strata and conormal directions not appearing in Λ_Σ . Therefore the remaining pieces are precisely polyhedral costandard sheaves compatible with the fan Σ , and these are generated by the sheaves $(i_{\sigma, \chi})_* \omega_{\chi + \sigma^\vee}$. $\square$

5.3 Theta Sheaves and Generation

We now introduce the basic constructible objects attached to the cones of Σ . For $\sigma \in \Sigma$, let $\sigma^\vee = \{m \in M_\mathbb{R} \mid \langle m, n \rangle \geq 0 \text{ for all } n \in \sigma\}$. For $\chi \in M_\sigma$, choose a lift of χ to M and set $P_{\sigma, \chi} := \chi + \sigma^\vee \subset M_\mathbb{R}$. This subset is independent of the chosen lift, since $\sigma^\perp$ is the lineality space of $\sigma^\vee$. Let $i_{\sigma, \chi} : P_{\sigma, \chi} \hookrightarrow M_\mathbb{R}$ be the inclusion. The constructible theta sheaf associated to (σ, χ) is

$$\Theta(\sigma, \chi) := (i_{\sigma, \chi})_* \omega_{P_{\sigma, \chi}}.$$

Thus $\Theta(\sigma, \chi)$ is the costandard sheaf supported on the translated dual cone $P_{\sigma, \chi}$. By [Proposition 5.4](#), one has $SS(\Theta(\sigma, \chi)) \subset \Lambda_\Sigma$, and hence $\Theta(\sigma, \chi) \in \text{Sh}_{\Lambda_\Sigma}(M_\mathbb{R})$.

Theorem 5.6 (FLTZ generation). *The category $\text{Sh}_{\Lambda_\Sigma}(M_\mathbb{R})$ is generated by the theta sheaves:*

$$\text{Sh}_{\Lambda_\Sigma}(M_\mathbb{R}) = \langle \Theta(\sigma, \chi) \mid \sigma \in \Sigma, \chi \in M_\sigma \rangle.$$

Here the right-hand side denotes the smallest stable idempotent-complete subcategory containing all $\Theta(\sigma, \chi)$.

Proof. Let $\mathcal{C}_\Sigma$ be the stable idempotent-complete subcategory generated by the theta sheaves. Since $SS(\Theta(\sigma, \chi)) \subset \Lambda_\Sigma$ for every $\sigma \in \Sigma$ and $\chi \in M_\sigma$, and since $\text{Sh}_{\Lambda_\Sigma}(M_\mathbb{R})$ is closed under shifts, cones, finite direct sums, and retracts, we have $\mathcal{C}_\Sigma \subset \text{Sh}_{\Lambda_\Sigma}(M_\mathbb{R})$.

Conversely, [Lemma 5.5](#) says that every object of $\text{Sh}_{\Lambda_\Sigma}(M_\mathbb{R})$ belongs to the stable idempotent-complete subcategory generated by the sheaves $(i_{\sigma, \chi})_* \omega_{\chi + \sigma^\vee} = \Theta(\sigma, \chi)$. Hence $\text{Sh}_{\Lambda_\Sigma}(M_\mathbb{R}) \subset \mathcal{C}_\Sigma$. The two inclusions give the desired equality. $\square$

6 The Toric Coherent–Constructible Correspondence

We now state the toric coherent–constructible correspondence. Let X_Σ be a smooth toric variety with acting torus T , and let $\Lambda_\Sigma \subset T^*M_\mathbb{R}$ be the FLTZ skeleton associated to the fan Σ . The coherent side of the compact form of the correspondence is the perfect equivariant category $\text{Perf}_T(X_\Sigma)$. In the proof, we first work with the theta-generated equivariant quasi-coherent category and then pass to compact objects, using [Corollary 4.8](#). The constructible side is the dg category of constructible sheaves on $M_\mathbb{R}$ with singular support contained in Λ_Σ .

Theorem 6.1 (Toric coherent–constructible correspondence). *Let X_Σ be a smooth toric variety. There is a quasi-equivalence of dg categories*

$$\kappa_\Sigma : \text{Perf}_T(X_\Sigma) \xrightarrow{\sim} \text{Sh}_{\Lambda_\Sigma}(M_\mathbb{R}).$$

Equivalently, on homotopy categories one obtains an equivalence

$$D^b(\text{Coh}_T(X_\Sigma)) \simeq H^0(\text{Sh}_{\Lambda_\Sigma}(M_\mathbb{R})).$$

The proof will be given by comparing the theta generators on the two sides. The objects $\Theta'(\sigma, \chi)$ and $\Theta(\sigma, \chi)$ are indexed by the same data (σ, χ) , and the functor κ_Σ is characterized by

$$\kappa_\Sigma(\Theta'(\sigma, \chi)) = \Theta(\sigma, \chi).$$

It remains to compare the Hom-complexes between these generators and then apply the generator criterion for quasi-equivalences.

6.1 The Correspondence of Generators

The theta objects on the two sides are indexed by the same combinatorial data. More precisely, for each cone $\sigma \in \Sigma$, let $q_\sigma : M \rightarrow M_\sigma$ be the quotient map.

Lemma 6.2. *Let $\sigma \in \Sigma$, and let $\chi, \chi' \in M$. If $q_\sigma(\chi) = q_\sigma(\chi')$, then $\Theta'(\sigma, \chi) \simeq \Theta'(\sigma, \chi')$. Hence the coherent-side theta object depends only on the class $\bar{\chi} \in M_\sigma$.*

Proof. The condition $q_\sigma(\chi) = q_\sigma(\chi')$ means that $\chi' - \chi \in \sigma^\perp \cap M$. Put $m_0 = \chi' - \chi$. Then the monomial χ^{m_0} is an invertible regular function on U_σ . Multiplication by χ^{m_0} gives a T -equivariant isomorphism $\mathcal{O}_{U_\sigma}(\chi) \simeq \mathcal{O}_{U_\sigma}(\chi')$, and pushing forward along $i_\sigma : U_\sigma \hookrightarrow X_\Sigma$ gives $\Theta'(\sigma, \chi) \simeq \Theta'(\sigma, \chi')$. $\square$

Thus we may write the coherent-side theta object as $\Theta'(\sigma, \bar{\chi})$, where $\bar{\chi} \in M_\sigma$. The constructible theta object $\Theta(\sigma, \bar{\chi})$ is already indexed by the same data. The correspondence of generators is therefore

$$\Theta'(\sigma, \bar{\chi}) \longmapsto \Theta(\sigma, \bar{\chi}).$$

The functor κ_Σ , once constructed, will be characterized on theta generators by

$$\kappa_\Sigma(\Theta'(\sigma, \bar{\chi})) = \Theta(\sigma, \bar{\chi}).$$

The next step is to compare the Hom-complexes between these generators. This comparison is the essential computation from which the CCC functor and the quasi-equivalence will follow.

6.2 Hom-Complex Computations

We now compare the Hom-complexes between theta objects. For $\bar{\chi} \in M_\sigma$, choose a lift $\chi \in M$ and write $P_{\sigma, \bar{\chi}} := \chi + \sigma^\vee$. This subset is independent of the chosen lift.

Lemma 6.3 (Hom between polyhedral costandard sheaves). *Let $P, Q \subset M_\mathbb{R}$ be closed convex polyhedra, and let $i_P : P \hookrightarrow M_\mathbb{R}$ and $i_Q : Q \hookrightarrow M_\mathbb{R}$ be the inclusions. Then*

$$\mathrm{RHom}((i_P)_*\omega_P, (i_Q)_*\omega_Q) \simeq \begin{cases} k, & P \subset Q, \\ 0, & \text{otherwise.} \end{cases}$$

In the nonzero case, the complex is concentrated in degree zero.

Proof. By Verdier duality, $(i_P)_*\omega_P \simeq D(k_P)$. Hence

$$\mathrm{RHom}((i_P)_*\omega_P, (i_Q)_*\omega_Q) \simeq \mathrm{RHom}(D(k_P), D(k_Q)) \simeq \mathrm{RHom}(k_Q, k_P).$$

For closed convex polyhedra, $\mathrm{RHom}(k_Q, k_P)$ is computed by the cohomology of the region $P \cap Q$ with the restriction condition from Q to P . If $P \subset Q$, this region is the convex polyhedron P , hence contractible, and the Hom-complex is k in degree zero. If $P \not\subset Q$, the corresponding restriction condition fails, and the Hom-complex is acyclic. This gives the stated formula. $\square$

Proposition 6.4 (Hom-complex comparison). *Let $\sigma, \tau \in \Sigma$, $\bar{\chi} \in M_\sigma$, and $\bar{\psi} \in M_\tau$. Then there is a natural quasi-isomorphism*

$$\mathrm{RHom}_T(\Theta'(\sigma, \bar{\chi}), \Theta'(\tau, \bar{\psi})) \simeq \mathrm{RHom}(\Theta(\sigma, \bar{\chi}), \Theta(\tau, \bar{\psi})).$$

More explicitly, after choosing lifts $\chi, \psi \in M$, both sides are quasi-isomorphic to

$$\begin{cases} k, & P_{\sigma, \bar{\chi}} \subset P_{\tau, \bar{\psi}}, \\ 0, & \text{otherwise.} \end{cases}$$

In the nonzero case, the complex is concentrated in degree zero.

Proof. We compute the two sides separately.

On the coherent side, write

$$\Theta'(\sigma, \bar{\chi}) = (i_\sigma)_*\mathcal{O}_{U_\sigma}(\chi), \quad \Theta'(\tau, \bar{\psi}) = (i_\tau)_*\mathcal{O}_{U_\tau}(\psi).$$

Since $U_\sigma \cap U_\tau = U_{\sigma \cap \tau}$ is again an affine toric chart, the Hom-complex is computed on affine charts by the corresponding graded module calculation. A T -equivariant morphism between the two equivariant line bundles is given by multiplication by a homogeneous monomial. With our linearization convention, the possible monomial has weight $\chi - \psi$.

This monomial defines a regular map on the relevant affine toric chart exactly when the corresponding character lies in the appropriate semigroup. In polyhedral terms, this condition is equivalent to

$$P_{\sigma, \bar{\chi}} \subset P_{\tau, \bar{\psi}}.$$

If this inclusion holds, the space of equivariant morphisms is one-dimensional, generated by this monomial. If the inclusion fails, there is no equivariant regular monomial map. Since the relevant toric intersections are affine, there are no higher derived Hom groups. Hence

$$\mathrm{RHom}_T(\Theta'(\sigma, \bar{\chi}), \Theta'(\tau, \bar{\psi})) \simeq \begin{cases} k, & P_{\sigma, \bar{\chi}} \subset P_{\tau, \bar{\psi}}, \\ 0, & \text{otherwise.} \end{cases}$$

On the constructible side, the theta objects are costandard sheaves on the closed convex polyhedra $P_{\sigma, \bar{\chi}}$ and $P_{\tau, \bar{\psi}}$:

$$\Theta(\sigma, \bar{\chi}) = (i_{\sigma, \bar{\chi}})_*\omega_{P_{\sigma, \bar{\chi}}}, \quad \Theta(\tau, \bar{\psi}) = (i_{\tau, \bar{\psi}})_*\omega_{P_{\tau, \bar{\psi}}}.$$

By Lemma 6.3, we get

$$\mathrm{RHom}(\Theta(\sigma, \bar{\chi}), \Theta(\tau, \bar{\psi})) \simeq \begin{cases} k, & P_{\sigma, \bar{\chi}} \subset P_{\tau, \bar{\psi}}, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore the coherent and constructible Hom-complexes are naturally quasi-isomorphic. The comparison is compatible with composition: if

$$P_{\sigma, \bar{\chi}} \subset P_{\tau, \bar{\psi}} \subset P_{\nu, \bar{\eta}},$$

then the generator of the first Hom-complex composes with the generator of the second to give the generator of the third. If one of the inclusions fails, the corresponding Hom-complex is zero. Hence the quasi-isomorphisms above are compatible with composition. $\square$

6.3 The CCC Functor

We now construct the theta-generated form of the CCC functor. Let $\mathcal{A}'_\Sigma$ be the full dg subcategory of $D(\mathrm{QCoh}_T(X_\Sigma))$ whose objects are the coherent-side theta objects $\Theta'(\sigma, \bar{\chi})$. Similarly, let $\mathcal{A}_\Sigma$ be the full dg subcategory of $\mathrm{Sh}_{\Lambda_\Sigma}(M_\mathbb{R})$ whose objects are the constructible theta objects $\Theta(\sigma, \bar{\chi})$.

Define a map on objects by

$$\Theta'(\sigma, \bar{\chi}) \mapsto \Theta(\sigma, \bar{\chi}).$$

By the Hom-complex comparison in [Proposition 6.4](#), for every pair of indices $(\sigma, \bar{\chi})$ and $(\tau, \bar{\psi})$, there is a natural quasi-isomorphism

$$\mathrm{RHom}_T(\Theta'(\sigma, \bar{\chi}), \Theta'(\tau, \bar{\psi})) \simeq \mathrm{RHom}(\Theta(\sigma, \bar{\chi}), \Theta(\tau, \bar{\psi})).$$

These quasi-isomorphisms are compatible with composition. Hence the assignment on theta objects extends to a dg functor

$$\kappa_\Sigma^0 : \mathcal{A}'_\Sigma \longrightarrow \mathcal{A}_\Sigma.$$

To obtain the theta-generated form of the CCC functor, we extend κ_Σ^0 to the triangulated dg subcategories generated by the theta objects. Since exact dg functors preserve shifts, cones, and finite direct sums, the assignment extends to an exact dg functor

$$\kappa_\Sigma : \langle \Theta'(\sigma, \bar{\chi}) \mid \sigma \in \Sigma, \bar{\chi} \in M_\sigma \rangle \longrightarrow \langle \Theta(\sigma, \bar{\chi}) \mid \sigma \in \Sigma, \bar{\chi} \in M_\sigma \rangle.$$

Equivalently, κ_Σ is the exact dg functor characterized by

$$\kappa_\Sigma(\Theta'(\sigma, \bar{\chi})) = \Theta(\sigma, \bar{\chi})$$

on theta generators, together with the Hom-complex identifications from [Proposition 6.4](#).

At this stage, κ_Σ is quasi-fully faithful on the theta generators. The compact coherent side will be obtained by passing to compact objects, as in [Corollary 4.8](#). The remaining step is to use the generation results to prove that the resulting functor is a quasi-equivalence.

6.4 Proof of the Quasi-Equivalence

We now prove that the functor constructed above gives the toric CCC. By construction, κ_Σ sends each coherent-side theta object to the corresponding constructible theta sheaf:

$$\kappa_\Sigma(\Theta'(\sigma, \bar{\chi})) = \Theta(\sigma, \bar{\chi}).$$

Moreover, [Proposition 6.4](#) shows that, for any $\sigma, \tau \in \Sigma$, $\bar{\chi} \in M_\sigma$, and $\bar{\psi} \in M_\tau$, the induced map on Hom-complexes

$$\mathrm{RHom}_T(\Theta'(\sigma, \bar{\chi}), \Theta'(\tau, \bar{\psi})) \longrightarrow \mathrm{RHom}(\Theta(\sigma, \bar{\chi}), \Theta(\tau, \bar{\psi}))$$

is a quasi-isomorphism.

It remains to use the generation results. On the coherent side, [Theorem 4.7](#) says that the theta objects generate the equivariant quasi-coherent derived category $D(\mathrm{QCoh}_T(X_\Sigma))$ as a localizing triangulated category. On the constructible side, [Theorem 5.6](#) says that the theta sheaves generate $\mathrm{Sh}_{\Lambda_\Sigma}(M_\mathbb{R})$. Hence the hypotheses of the generator criterion [Theorem 3.1](#) are satisfied for the theta-generated categories: the functor is exact, it identifies the Hom-complexes between the generators, and the corresponding theta objects generate the two sides.

Therefore κ_Σ gives a quasi-equivalence between the theta-generated coherent and constructible categories. Passing to compact objects on the coherent side, [Corollary 4.8](#) identifies the compact part of $D(\mathrm{QCoh}_T(X_\Sigma))$ with $\mathrm{Perf}_T(X_\Sigma)$. Thus we obtain the perfect form of the toric coherent–constructible correspondence:

$$\kappa_\Sigma : \mathrm{Perf}_T(X_\Sigma) \xrightarrow{\sim} \mathrm{Sh}_{\Lambda_\Sigma}(M_\mathbb{R}).$$

Since X_Σ is smooth, $\mathrm{Perf}_T(X_\Sigma)$ is a dg enhancement of $D^b(\mathrm{Coh}_T(X_\Sigma))$. Hence, on homotopy categories, one obtains

$$D^b(\mathrm{Coh}_T(X_\Sigma)) \simeq H^0(\mathrm{Sh}_{\Lambda_\Sigma}(M_\mathbb{R})).$$

This proves the toric coherent–constructible correspondence.

7 Fukaya-Theoretic Interpretation

The coherent–constructible correspondence proved in the previous section was formulated entirely in terms of coherent sheaves and constructible sheaves. The proof used the FLTZ skeleton as a singular-support condition and avoided any explicit use of Fukaya categories. Nevertheless, the constructible side has a natural symplectic interpretation: constructible sheaves on $M_\mathbb{R}$ with prescribed singular support may be regarded as a sheaf-theoretic model for a Fukaya category of the cotangent bundle $T^*M_\mathbb{R}$.

The purpose of this section is to explain this interpretation. We will not construct the Fukaya category from first principles. Instead, we use the microlocal sheaf–Fukaya correspondence as a guiding principle: a conic Lagrangian subset of a cotangent bundle controls both singular supports of constructible sheaves and asymptotic supports of Lagrangian branes. In our case, the relevant conic Lagrangian is precisely the FLTZ skeleton Λ_Σ .

Thus the toric CCC may be viewed as part of a larger triangle of correspondences:

$$\mathrm{Perf}_T(X_\Sigma) \simeq \mathrm{Sh}_{\Lambda_\Sigma}(M_\mathbb{R}) \simeq \mathcal{W}(T^*M_\mathbb{R}, \Lambda_\Sigma^\infty),$$

where the right-hand side denotes a wrapped or partially wrapped Fukaya category with stop determined by the contact boundary Λ_Σ^∞ of the FLTZ skeleton. This equivalence should be understood up to the usual conventions concerning compact objects, brane structures, and possible opposite categories.

In this sense, the constructible side is not merely a combinatorial replacement for coherent sheaves. It is also a microlocal model for the Fukaya-theoretic mirror of the toric variety. The following subsections explain how the singular-support condition becomes a Lagrangian support condition, how microlocal sheaves are related to wrapped Fukaya categories, and how the theta sheaves appearing in the CCC may be interpreted as Lagrangian branes.

7.1 From Singular Support to Lagrangian Support

In Section 5, the FLTZ skeleton $\Lambda_\Sigma \subset T^*M_\mathbb{R}$ was introduced as the singular-support condition defining the constructible side of the CCC. The purpose of this subsection is to explain how the same subset is interpreted on the Fukaya-theoretic side.

Definition 7.1 (Cotangent symplectic structure). Let Y be a smooth real manifold. The cotangent bundle T^*Y carries its canonical Liouville one-form λ . Its exterior derivative

$$\omega = d\lambda$$

is the canonical exact symplectic form on T^*Y .

Definition 7.2 (Lagrangian submanifold). Let (X, ω) be a symplectic manifold. A submanifold $L \subset X$ is called Lagrangian if $\omega|_L = 0$ and $\dim L = \frac{1}{2} \dim X$. In particular, a Lagrangian submanifold of T^*Y has dimension $\dim Y$.

Definition 7.3 (Lagrangian brane). In Fukaya theory, one usually works not merely with Lagrangian submanifolds, but with Lagrangians equipped with additional structures such as orientations, gradings, spin structures, and local systems. Such an enhanced Lagrangian is called a Lagrangian brane. In this section, the word “brane” will be used only in this sense.

Definition 7.4 (Boundary at infinity of a conic subset). Let $\Lambda \subset T^*Y$ be a conic subset, namely a subset invariant under positive scaling in the cotangent fibers. Its boundary at infinity is

$$\Lambda^\infty := (\Lambda \setminus T_Y^*Y) / \mathbb{R}_{>0} \subset S^*Y,$$

where T_Y^*Y denotes the zero section and

$$S^*Y := (T^*Y \setminus T_Y^*Y) / \mathbb{R}_{>0}$$

is the cosphere bundle.

For $Y = M_\mathbb{R}$, we have the identification $T^*M_\mathbb{R} \simeq M_\mathbb{R} \times N_\mathbb{R}$. Under this identification, the fiberwise scaling action is the scaling action on the $N_\mathbb{R}$ -factor. By [Proposition 5.2](#), the FLTZ skeleton Λ_Σ is conic. Hence its boundary at infinity

$$\Lambda_\Sigma^\infty := (\Lambda_\Sigma \setminus T_{M_\mathbb{R}}^*M_\mathbb{R}) / \mathbb{R}_{>0} \subset S^*M_\mathbb{R}$$

is well-defined.

On the sheaf-theoretic side, a constructible sheaf F has singular support $SS(F) \subset T^*M_\mathbb{R}$. This subset records the cotangent directions in which local sections of F fail to propagate. Therefore the condition

$$SS(F) \subset \Lambda_\Sigma$$

means that F is allowed to have singular behavior only in the microlocal directions prescribed by the FLTZ skeleton.

On the Fukaya-theoretic side, the analogous restriction is imposed on Lagrangian branes in $T^*M_\mathbb{R}$. The subset $\Lambda_\Sigma^\infty \subset S^*M_\mathbb{R}$ records the directions at infinity determined by the FLTZ skeleton. In wrapped Fukaya theory, such a subset is used as a stop: it restricts the allowed wrapping behavior of Lagrangian branes at infinity.

Thus the same geometric object has two interpretations. On the constructible side, Λ_Σ is a singular-support constraint. On the Fukaya side, Λ_Σ^∞ is an asymptotic Lagrangian constraint. This is the basic bridge from constructible sheaves to Fukaya categories; the precise categorical form of this bridge will be recalled in the next subsection.

7.2 Microlocal Sheaves and Wrapped Fukaya Categories

We now recall the external theorem relating constructible sheaves to Fukaya categories. The terminology concerning conic subsets, their boundary at infinity, and stops was fixed in the previous subsection. This theorem is not proved in this paper; it is used only to interpret the constructible side of the CCC as a Fukaya-theoretic category.

Let Y be a real analytic manifold, and let $\Lambda \subset T^*Y$ be a closed conic subanalytic Lagrangian subset. We write $Sh_\Lambda(Y; k)^c$ for the compact objects in the dg category of constructible k -sheaves on Y whose singular support is contained in Λ . We write $\mathcal{W}(T^*Y, \Lambda^\infty; k)$ for the partially wrapped Fukaya category of T^*Y stopped at Λ^∞ , with the brane and grading conventions fixed.

Theorem 7.5 (Microlocal sheaf–Fukaya correspondence). *With the notation above, there is an equivalence of dg categories*

$$Sh_\Lambda(Y; k)^c \simeq \mathcal{W}(T^*Y, \Lambda^\infty; k).$$

We use this theorem as an external input. Depending on the convention for Fukaya morphisms and for standard versus costandard sheaves, the right-hand side may have to be replaced by its opposite category. This convention does not affect the role of the theorem here.

Applying the theorem to $Y = M_\mathbb{R}$ and $\Lambda = \Lambda_\Sigma$, we obtain

$$Sh_{\Lambda_\Sigma}(M_\mathbb{R}; k)^c \simeq \mathcal{W}(T^*M_\mathbb{R}, \Lambda_\Sigma^\infty; k).$$

Thus the constructible side of the toric CCC may be viewed as a microlocal sheaf model for the stopped wrapped Fukaya category determined by the FLTZ skeleton.

7.3 Theta Sheaves as Branes

We now apply the microlocal sheaf–Fukaya correspondence from the previous subsection to the theta sheaves appearing in the CCC. Let

$$\Phi_\Sigma : Sh_{\Lambda_\Sigma}(M_\mathbb{R}; k)^c \xrightarrow{\sim} \mathcal{W}(T^*M_\mathbb{R}, \Lambda_\Sigma^\infty; k)$$

be an equivalence as in [Theorem 7.5](#). We use this equivalence to transport the constructible theta objects to the Fukaya side.

Lemma 7.6. *For every $\sigma \in \Sigma$ and $\bar{\chi} \in M_\sigma$, the theta sheaf $\Theta(\sigma, \bar{\chi})$ has singular support contained in Λ_Σ . Hence it defines an object of $Sh_{\Lambda_\Sigma}(M_\mathbb{R}; k)$.*

Proof. By definition, $\Theta(\sigma, \bar{\chi})$ is the costandard sheaf supported on the closed convex polyhedron $P_{\sigma, \bar{\chi}} = \bar{\chi} + \sigma^\vee$. The singular support of such a costandard polyhedral sheaf is controlled by the conormal cones to the faces of $P_{\sigma, \bar{\chi}}$. These conormal directions are precisely the directions appearing in the FLTZ skeleton. Therefore $SS(\Theta(\sigma, \bar{\chi})) \subset \Lambda_\Sigma$. $\square$

Definition 7.7. The elementary theta brane associated to $(\sigma, \bar{\chi})$ is the Fukaya object

$$L_{\sigma, \bar{\chi}} := \Phi_\Sigma(\Theta(\sigma, \bar{\chi})) \in \mathcal{W}(T^*M_\mathbb{R}, \Lambda_\Sigma^\infty; k).$$

Thus $L_{\sigma, \bar{\chi}}$ is not introduced by an independent Floer-theoretic construction in this paper. It is defined as the Fukaya object corresponding to the theta sheaf under the microlocal sheaf–Fukaya equivalence. The polyhedron $P_{\sigma, \bar{\chi}}$ records the base support of the sheaf, while the conormal geometry of its faces records the Lagrangian directions controlled by the FLTZ skeleton.

Proposition 7.8 (Morphisms between elementary theta branes). *Let $\sigma, \tau \in \Sigma$, $\bar{\chi} \in M_\sigma$, and $\bar{\psi} \in M_\tau$. Then*

$$\mathrm{RHom}_\mathcal{W}(L_{\sigma, \bar{\chi}}, L_{\tau, \bar{\psi}}) \simeq \mathrm{RHom}(\Theta(\sigma, \bar{\chi}), \Theta(\tau, \bar{\psi})).$$

Consequently,

$$\mathrm{RHom}_\mathcal{W}(L_{\sigma, \bar{\chi}}, L_{\tau, \bar{\psi}}) \simeq \begin{cases} k, & P_{\sigma, \bar{\chi}} \subset P_{\tau, \bar{\psi}}, \\ 0, & \text{otherwise.} \end{cases}$$

In the nonzero case, the complex is concentrated in degree zero.

Proof. The first quasi-isomorphism follows from the fact that Φ_Σ is an equivalence of dg categories. The explicit computation then follows from the Hom-complex calculation for constructible theta sheaves proved in [Proposition 6.4](#). $\square$

Thus the theta sheaves become elementary Fukaya branes under the microlocal sheaf–Fukaya correspondence. In the sheaf-theoretic proof of the CCC, the objects $\Theta(\sigma, \bar{\chi})$ generate the constructible side. In the Fukaya-theoretic interpretation, the corresponding objects $L_{\sigma, \bar{\chi}}$ form the elementary branes whose asymptotic behavior is governed by Λ_Σ^∞ .

7.4 The Fukaya-Theoretic Form of the CCC

We now combine the toric CCC with the microlocal sheaf–Fukaya correspondence recalled from [Theorem 7.5](#). The CCC proved in [Theorem 6.1](#) gives an equivalence between the equivariant coherent category of the toric variety and the constructible category with singular support in the FLTZ skeleton. The microlocal sheaf–Fukaya correspondence identifies this constructible category with a stopped wrapped Fukaya category of the cotangent bundle.

Corollary 7.9 (Fukaya-theoretic form of the toric CCC). *Let X_Σ be a smooth toric variety, and let $\Lambda_\Sigma \subset T^*M_\mathbb{R}$ be the FLTZ skeleton. After fixing the Fukaya-theoretic brane and grading conventions, there is an equivalence of dg categories*

$$\mathrm{Perf}_T(X_\Sigma) \simeq \mathcal{W}(T^*M_\mathbb{R}, \Lambda_\Sigma^\infty; k).$$

Equivalently, one has a chain of equivalences

$$\mathrm{Perf}_T(X_\Sigma) \simeq \mathrm{Sh}_{\Lambda_\Sigma}(M_\mathbb{R}; k)^c \simeq \mathcal{W}(T^*M_\mathbb{R}, \Lambda_\Sigma^\infty; k).$$

Proof. The first equivalence in the chain is the compact form of the toric coherent–constructible correspondence proved from [Theorem 6.1](#). The second equivalence is the microlocal sheaf–Fukaya correspondence from [Theorem 7.5](#), applied to $Y = M_\mathbb{R}$ and $\Lambda = \Lambda_\Sigma$. Composing these equivalences gives the desired Fukaya-theoretic form of the CCC. $\square$

Thus the equivariant coherent category of the toric variety can be interpreted as a stopped wrapped Fukaya category of the cotangent bundle $T^*M_\mathbb{R}$. Under this interpretation, the FLTZ skeleton plays the role of the stop at infinity. On the constructible side, it appears as the singular-support condition

$$SS(F) \subset \Lambda_\Sigma.$$

On the Fukaya side, its boundary at infinity $\Lambda_\Sigma^\infty \subset S^*M_\mathbb{R}$ controls the allowed asymptotic behavior of Lagrangian branes.

The theta objects also admit a Fukaya-theoretic interpretation. The coherent theta object $\Theta'(\sigma, \bar{\chi})$ corresponds under the CCC to the constructible theta sheaf $\Theta(\sigma, \bar{\chi})$, and under the microlocal sheaf–Fukaya correspondence this object corresponds to the elementary theta brane $L_{\sigma, \bar{\chi}}$. Therefore the correspondence on generators can be summarized as

$$\Theta'(\sigma, \bar{\chi}) \longleftrightarrow \Theta(\sigma, \bar{\chi}) \longleftrightarrow L_{\sigma, \bar{\chi}}.$$

In this way, the Fukaya-theoretic form of the CCC identifies coherent theta objects with elementary Lagrangian branes governed by the same polyhedral data.

This formulation explains the role of the symplectic material from the beginning of the paper. The moment map and moment polytope describe how convex geometry arises naturally from toric symplectic geometry, while the FLTZ skeleton gives the corresponding microlocal Lagrangian support condition inside $T^*M_\mathbb{R}$. The sheaf-theoretic CCC can therefore be viewed as a bridge between equivariant coherent sheaves on X_Σ and Fukaya objects controlled by the FLTZ skeleton.

8 Examples of the Toric CCC

The preceding sections established the toric coherent–constructible correspondence in its general form. The proof was based on theta generators, the FLTZ skeleton, and the comparison of Hom-complexes. In this section we illustrate these constructions in several explicit examples.

In each example, we compute the same basic objects. We first describe the fan Σ and the associated toric variety X_Σ . We then write down the FLTZ skeleton

$$\Lambda_\Sigma = \bigcup_{\sigma \in \Sigma} \bigcup_{\chi \in M_\sigma} (\chi + \sigma^\perp) \times (-\sigma) \subset T^*M_\mathbb{R}.$$

This specifies the allowed singular-support directions on the constructible side.

Next, for each cone $\sigma \in \Sigma$ and character class $\chi \in M_\sigma$, we identify the constructible theta sheaf

$$\Theta(\sigma, \chi) = (i_{\chi + \sigma^\vee})_* \omega_{\chi + \sigma^\vee}.$$

We compare it with the coherent theta object

$$\Theta'(\sigma, \chi) = (i_\sigma)_* \mathcal{O}_{U_\sigma}(\chi).$$

Thus both sides are indexed by the same combinatorial data (σ, χ) .

The main calculation is the Hom-complex between theta objects. On the constructible side, this is determined by an inclusion of polyhedra:

$$\mathrm{RHom}(\Theta(\sigma, \chi), \Theta(\tau, \psi)) \simeq \begin{cases} k, & \chi + \sigma^\vee \subset \psi + \tau^\vee, \\ 0, & \text{otherwise.} \end{cases}$$

The same condition appears on the coherent side as the regularity condition for an equivariant monomial map. Therefore the examples make explicit the basic dictionary

$$\text{equivariant monomial maps} \longleftrightarrow \text{inclusions of polyhedral sheaves.}$$

8.1 The affine line $\mathbb{A}^1$

We begin with the affine line. This is the simplest example in which the FLTZ skeleton has a nontrivial conormal direction and the theta sheaves are supported on half-lines.

Let $N = M = \mathbb{Z}$, and identify $N_{\mathbb{R}}$ and $M_{\mathbb{R}}$ with $\mathbb{R}$. Let Σ be the fan consisting of the zero cone 0 and the ray $\rho = \mathbb{R}_{\geq 0}$. The corresponding toric variety is $X_{\Sigma} = \mathbb{A}^1$. Under the identification $T^*M_{\mathbb{R}} \simeq M_{\mathbb{R}} \times N_{\mathbb{R}}$, the zero cone contributes the zero section $\mathbb{R} \times \{0\}$. For the ray ρ , one has $\rho^{\perp} = 0$, $M_{\rho} = M = \mathbb{Z}$, and $-\rho = \mathbb{R}_{\leq 0}$. Therefore

$$\Lambda_{\Sigma} = \mathbb{R} \times \{0\} \cup \bigcup_{a \in \mathbb{Z}} \{a\} \times \mathbb{R}_{\leq 0}.$$

Thus a sheaf in $Sh_{\Lambda_{\Sigma}}(\mathbb{R})$ may have singularities only at integral points, and only in the negative cotangent direction.

We now describe the theta sheaves. The zero cone gives $\sigma^{\vee} = M_{\mathbb{R}} = \mathbb{R}$, so the corresponding constructible theta sheaf is the costandard sheaf on the whole line:

$$\Theta(0, 0) = \omega_{\mathbb{R}}.$$

For the ray ρ , we have $\rho^{\vee} = \mathbb{R}_{\geq 0}$. Hence, for $a \in M_{\rho} = \mathbb{Z}$, the associated polyhedron is

$$P_{\rho, a} = a + \rho^{\vee} = [a, \infty),$$

and the constructible theta sheaf is

$$\Theta(\rho, a) = (i_{[a, \infty)})_* \omega_{[a, \infty)}.$$

Thus the constructible theta objects for $\mathbb{A}^1$ are the costandard sheaves on the half-lines $[a, \infty)$, together with the costandard sheaf on $\mathbb{R}$ coming from the zero cone.

On the coherent side, the affine chart corresponding to ρ is $U_{\rho} = \mathbb{A}^1$, with coordinate ring $k[x]$ and grading $\deg x = 1$. The coherent theta object attached to (ρ, a) is

$$\Theta'(\rho, a) = \mathcal{O}_{\mathbb{A}^1}(a),$$

where the underlying line bundle is trivial and the equivariant structure has weight a . The zero cone corresponds to the dense torus $U_0 = \mathbb{G}_m$; the associated theta object is $\Theta'(0, 0) = (j_0)_* \mathcal{O}_{\mathbb{G}_m}$, where $j_0 : \mathbb{G}_m \hookrightarrow \mathbb{A}^1$ is the open embedding. The correspondence sends

$$\Theta'(\rho, a) \longmapsto \Theta(\rho, a), \quad \Theta'(0, 0) \longmapsto \Theta(0, 0).$$

The Hom-complexes are determined by inclusions of the corresponding polyhedra. For $a, b \in \mathbb{Z}$, one has

$$\mathrm{RHom}(\Theta(\rho, a), \Theta(\rho, b)) \simeq \begin{cases} k, & [a, \infty) \subset [b, \infty), \\ 0, & \text{otherwise.} \end{cases}$$

Since $[a, \infty) \subset [b, \infty)$ if and only if $a \geq b$, this becomes

$$\mathrm{RHom}(\Theta(\rho, a), \Theta(\rho, b)) \simeq \begin{cases} k, & a \geq b, \\ 0, & a < b. \end{cases}$$

The other Hom-complexes involving the zero-cone object are also given by the same inclusion rule:

$$\mathrm{RHom}(\Theta(0, 0), \Theta(\rho, a)) \simeq 0, \quad \mathrm{RHom}(\Theta(\rho, a), \Theta(0, 0)) \simeq k.$$

Indeed, $\mathbb{R} \not\subset [a, \infty)$, while $[a, \infty) \subset \mathbb{R}$.

This calculation agrees with the coherent side. A T -equivariant morphism $\mathcal{O}_{\mathbb{A}^1}(a) \rightarrow \mathcal{O}_{\mathbb{A}^1}(b)$ is given by multiplication by a monomial of weight $a - b$. Such a monomial is regular on $\mathbb{A}^1$ precisely when $a - b \geq 0$, or equivalently $a \geq b$. Hence

$$\mathrm{RHom}_T(\Theta'(\rho, a), \Theta'(\rho, b)) \simeq \mathrm{RHom}(\Theta(\rho, a), \Theta(\rho, b)).$$

Moreover, if $a \geq b \geq c$, the composition of the maps corresponding to $a \geq b$ and $b \geq c$ is the map corresponding to $a \geq c$. On the coherent side this is the equality $x^{a-b}x^{b-c} = x^{a-c}$, while on the constructible side it is composition of the inclusion maps between the corresponding half-line sheaves.

Thus, in the affine-line case, the coherent-constructible correspondence reduces to the elementary dictionary

$$\text{regular monomial } x^{a-b} \longleftrightarrow [a, \infty) \subset [b, \infty).$$

8.2 The projective plane $\mathbb{P}^2$

We now consider the first two-dimensional projective example. Let $N = M = \mathbb{Z}^2$, and let Σ be the complete fan in $N_{\mathbb{R}}$ with rays generated by

$$v_1 = (1, 0), \quad v_2 = (0, 1), \quad v_3 = (-1, -1).$$

The maximal cones are

$$\sigma_{12} = \langle v_1, v_2 \rangle, \quad \sigma_{23} = \langle v_2, v_3 \rangle, \quad \sigma_{31} = \langle v_3, v_1 \rangle.$$

The corresponding toric variety is $X_{\Sigma} = \mathbb{P}^2$. We write a point of $M_{\mathbb{R}}$ as $m = (m_1, m_2)$, and identify $T^*M_{\mathbb{R}}$ with $M_{\mathbb{R}} \times N_{\mathbb{R}}$.

We first describe the FLTZ skeleton. The zero cone contributes the zero section $M_{\mathbb{R}} \times \{0\}$. The three rays contribute conormal directions along three families of integral affine lines. More explicitly, for $\rho_1 = \mathbb{R}_{\geq 0}v_1$, one has $\rho_1^{\perp} = \{m_1 = 0\}$, and the quotient M_{ρ_1} is indexed by the integer value of m_1 . Hence ρ_1 contributes

$$\bigcup_{a \in \mathbb{Z}} \{m_1 = a\} \times \mathbb{R}_{\leq 0}v_1.$$

Similarly, $\rho_2 = \mathbb{R}_{\geq 0}v_2$ contributes

$$\bigcup_{b \in \mathbb{Z}} \{m_2 = b\} \times \mathbb{R}_{\leq 0}v_2.$$

For $\rho_3 = \mathbb{R}_{\geq 0}v_3$, the equation for $\rho_3^{\perp}$ is $m_1 + m_2 = 0$. Thus ρ_3 contributes

$$\bigcup_{c \in \mathbb{Z}} \{m_1 + m_2 = c\} \times \mathbb{R}_{\geq 0}(1, 1),$$

because $-\rho_3 = \mathbb{R}_{\geq 0}(1, 1)$.

The maximal cones contribute conormal directions over integral lattice points. Since each maximal cone has $\sigma^{\perp} = 0$, the quotient M_{σ} is just M . Therefore

$$\Lambda_{\Sigma} = M_{\mathbb{R}} \times \{0\} \cup \bigcup_{i=1}^3 \bigcup_{\chi \in M_{\rho_i}} (\chi + \rho_i^{\perp}) \times (-\rho_i) \cup \bigcup_{\sigma \in \Sigma(2)} \bigcup_{\chi \in M} \{\chi\} \times (-\sigma).$$

Geometrically, the base of this skeleton is controlled by the integral affine arrangement consisting of the lines

$$m_1 = a, \quad m_2 = b, \quad m_1 + m_2 = c,$$

where $a, b, c \in \mathbb{Z}$. The cotangent directions are prescribed by the corresponding cones of the fan.

We now compute the constructible theta sheaves. The zero cone gives

$$\Theta(0, 0) = \omega_{M_{\mathbb{R}}}.$$

For the ray ρ_1 , the dual cone is $\rho_1^{\vee} = \{m_1 \geq 0\}$. Hence, for $a \in \mathbb{Z}$,

$$P_{\rho_1, a} = \{m_1 \geq a\}, \quad \Theta(\rho_1, a) = (i_{\{m_1 \geq a\}})_* \omega_{\{m_1 \geq a\}}.$$

Similarly,

$$P_{\rho_2, b} = \{m_2 \geq b\}, \quad \Theta(\rho_2, b) = (i_{\{m_2 \geq b\}})_* \omega_{\{m_2 \geq b\}},$$

and

$$P_{\rho_3, c} = \{m_1 + m_2 \leq c\}, \quad \Theta(\rho_3, c) = (i_{\{m_1 + m_2 \leq c\}})_* \omega_{\{m_1 + m_2 \leq c\}}.$$

Thus the ray generators give costandard sheaves on three types of half-planes.

For the maximal cone σ_{12} , one has

$$\sigma_{12}^{\vee} = \{m_1 \geq 0, m_2 \geq 0\}.$$

Therefore, for $\chi = (a, b) \in M$,

$$P_{\sigma_{12}, \chi} = \{m_1 \geq a, m_2 \geq b\}.$$

This is a translated quadrant in $M_{\mathbb{R}}$, and

$$\Theta(\sigma_{12}, \chi) = (i_{P_{\sigma_{12}, \chi}})_* \omega_{P_{\sigma_{12}, \chi}}.$$

For the other two maximal cones, the corresponding polyhedra are

$$P_{\sigma_{23}, (a, b)} = \{m_2 \geq b, m_1 + m_2 \leq a + b\},$$

and

$$P_{\sigma_{31}, (a, b)} = \{m_1 \geq a, m_1 + m_2 \leq a + b\}.$$

Thus the maximal cones give costandard sheaves on translated two-dimensional sectors.

On the coherent side, the affine chart $U_{\sigma_{12}}$ is isomorphic to $\mathbb{A}^2$, with coordinate ring

$$\mathbb{C}[\sigma_{12}^\vee \cap M] \cong \mathbb{C}[x, y].$$

The theta object associated to $(\sigma_{12}, (a, b))$ is

$$\Theta'(\sigma_{12}, (a, b)) = (i_{\sigma_{12}})_* \mathcal{O}_{U_{\sigma_{12}}}(a, b),$$

where the equivariant structure has weight (a, b) . The same description applies to the other affine charts. Under the correspondence,

$$\Theta'(\sigma, \chi) \longmapsto \Theta(\sigma, \chi)$$

for every cone $\sigma \in \Sigma$ and every $\chi \in M_\sigma$.

Let us spell out the Hom-computation for the cone σ_{12} . For $\chi = (a, b)$ and $\psi = (a', b')$, we have

$$P_{\sigma_{12}, \chi} \subset P_{\sigma_{12}, \psi}$$

if and only if

$$a \geq a', \quad b \geq b'.$$

Therefore

$$\mathrm{RHom}(\Theta(\sigma_{12}, (a, b)), \Theta(\sigma_{12}, (a', b'))) \simeq \begin{cases} k, & a \geq a' \text{ and } b \geq b', \\ 0, & \text{otherwise.} \end{cases}$$

This agrees with the coherent calculation. A T -equivariant morphism

$$\mathcal{O}_{U_{\sigma_{12}}}(a, b) \longrightarrow \mathcal{O}_{U_{\sigma_{12}}}(a', b')$$

is given by multiplication by a monomial of weight $(a - a', b - b')$. This monomial is regular on $U_{\sigma_{12}} \cong \mathbb{A}^2$ precisely when

$$(a - a', b - b') \in \sigma_{12}^\vee \cap M,$$

which is equivalent to $a \geq a'$ and $b \geq b'$. Hence

$$\mathrm{RHom}_T(\Theta'(\sigma_{12}, (a, b)), \Theta'(\sigma_{12}, (a', b'))) \simeq \mathrm{RHom}(\Theta(\sigma_{12}, (a, b)), \Theta(\sigma_{12}, (a', b'))).$$

The same computation holds for the other cones, with the corresponding dual cone replacing $\sigma_{12}^\vee$. In general, the Hom-complex is nonzero exactly when

$$\chi + \sigma^\vee \subset \psi + \tau^\vee.$$

Thus, for $\mathbb{P}^2$, the coherent side records regular equivariant monomial maps on the affine toric charts, while the constructible side records inclusions among half-planes and sectors in $M_\mathbb{R}$.

It is also useful to record how familiar line bundles appear in this example. Let $D = dD_{\rho_3}$, with $d \geq 0$. Then the associated polytope is

$$P_D = \{(m_1, m_2) \in M_\mathbb{R} \mid m_1 \geq 0, m_2 \geq 0, m_1 + m_2 \leq d\}.$$

This is the standard triangle of side length d . Under the correspondence, the equivariant line bundle $\mathcal{O}_{\mathbb{P}^2}(D) \cong \mathcal{O}_{\mathbb{P}^2}(d)$ is represented on the constructible side by the costandard sheaf supported on this triangle:

$$\kappa_\Sigma(\mathcal{O}_{\mathbb{P}^2}(d)) \simeq (i_{P_D})_* \omega_{P_D}.$$

Thus the usual monomial basis of $H^0(\mathbb{P}^2, \mathcal{O}(d))$ is reflected by the lattice points of the triangular support P_D . In this way, the projective-plane example makes visible the passage from equivariant coherent data to polyhedral constructible sheaves.

8.3 The product $\mathbb{P}^1 \times \mathbb{P}^1$

We next consider the product $\mathbb{P}^1 \times \mathbb{P}^1$. This example is useful because its fan is the product of the fan of $\mathbb{P}^1$ with itself, and the corresponding polyhedral picture is rectangular rather than triangular.

Let $N = M = \mathbb{Z}^2$, with coordinates $m = (m_1, m_2)$ on $M_\mathbb{R}$. The fan Σ has four rays generated by

$$v_1 = (1, 0), \quad v_2 = (0, 1), \quad v_3 = (-1, 0), \quad v_4 = (0, -1),$$

and four maximal cones

$$\sigma_{12} = \langle v_1, v_2 \rangle, \quad \sigma_{23} = \langle v_2, v_3 \rangle, \quad \sigma_{34} = \langle v_3, v_4 \rangle, \quad \sigma_{41} = \langle v_4, v_1 \rangle.$$

The associated toric variety is $X_\Sigma = \mathbb{P}^1 \times \mathbb{P}^1$.

We first describe the FLTZ skeleton. The zero cone contributes the zero section $M_{\mathbb{R}} \times \{0\}$. The rays $\rho_1 = \mathbb{R}_{\geq 0}v_1$ and $\rho_3 = \mathbb{R}_{\geq 0}v_3$ both have annihilator $\{m_1 = 0\}$, so they contribute conormal directions along the vertical integral lines $m_1 = a$. More precisely,

$$\rho_1 : \bigcup_{a \in \mathbb{Z}} \{m_1 = a\} \times \mathbb{R}_{\leq 0}v_1, \quad \rho_3 : \bigcup_{a \in \mathbb{Z}} \{m_1 = a\} \times \mathbb{R}_{\geq 0}v_1.$$

Similarly, the rays $\rho_2 = \mathbb{R}_{\geq 0}v_2$ and $\rho_4 = \mathbb{R}_{\geq 0}v_4$ contribute conormal directions along the horizontal integral lines $m_2 = b$:

$$\rho_2 : \bigcup_{b \in \mathbb{Z}} \{m_2 = b\} \times \mathbb{R}_{\leq 0}v_2, \quad \rho_4 : \bigcup_{b \in \mathbb{Z}} \{m_2 = b\} \times \mathbb{R}_{\geq 0}v_2.$$

The maximal cones contribute conic directions over the lattice points of M . Thus

$$\Lambda_{\Sigma} = M_{\mathbb{R}} \times \{0\} \cup \bigcup_{\rho \in \Sigma(1)} \bigcup_{\chi \in M_{\rho}} (\chi + \rho^{\perp}) \times (-\rho) \cup \bigcup_{\sigma \in \Sigma(2)} \bigcup_{\chi \in M} \{\chi\} \times (-\sigma).$$

Geometrically, the base of the skeleton is controlled by the integral grid consisting of the vertical and horizontal lines

$$m_1 = a, \quad m_2 = b, \quad a, b \in \mathbb{Z}.$$

This contrasts with the case of $\mathbb{P}^2$, where a third diagonal family of affine lines also appears.

We now identify the theta sheaves. The zero cone again gives

$$\Theta(0, 0) = \omega_{M_{\mathbb{R}}}.$$

For the four rays, the dual cones are half-spaces. Hence the corresponding theta sheaves are costandard sheaves on vertical or horizontal half-planes:

$$\Theta(\rho_1, a) = (i_{\{m_1 \geq a\}})_* \omega_{\{m_1 \geq a\}}, \quad \Theta(\rho_3, a) = (i_{\{m_1 \leq a\}})_* \omega_{\{m_1 \leq a\}},$$

and

$$\Theta(\rho_2, b) = (i_{\{m_2 \geq b\}})_* \omega_{\{m_2 \geq b\}}, \quad \Theta(\rho_4, b) = (i_{\{m_2 \leq b\}})_* \omega_{\{m_2 \leq b\}}.$$

For the maximal cones, the theta sheaves are costandard sheaves on translated quadrants. If $\chi = (a, b) \in M$, then

$$P_{\sigma_{12}, \chi} = \{m_1 \geq a, m_2 \geq b\},$$

$$P_{\sigma_{23}, \chi} = \{m_1 \leq a, m_2 \geq b\},$$

$$P_{\sigma_{34}, \chi} = \{m_1 \leq a, m_2 \leq b\},$$

and

$$P_{\sigma_{41}, \chi} = \{m_1 \geq a, m_2 \leq b\}.$$

Thus each maximal cone gives one of the four possible quadrant directions in $M_{\mathbb{R}}$, and

$$\Theta(\sigma, \chi) = (i_{P_{\sigma, \chi}})_* \omega_{P_{\sigma, \chi}}.$$

On the coherent side, each maximal cone gives an affine chart isomorphic to $\mathbb{A}^2$. For example,

$$U_{\sigma_{12}} = \text{Spec } \mathbb{C}[\sigma_{12}^{\vee} \cap M] \cong \mathbb{A}^2,$$

with coordinate ring $\mathbb{C}[x, y]$, where $\deg x = (1, 0)$ and $\deg y = (0, 1)$. The coherent theta object attached to $(\sigma_{12}, (a, b))$ is

$$\Theta'(\sigma_{12}, (a, b)) = (i_{\sigma_{12}})_* \mathcal{O}_{U_{\sigma_{12}}}(a, b).$$

The other affine charts are treated in the same way, using the corresponding semigroup algebras. Under the correspondence,

$$\Theta'(\sigma, \chi) \longmapsto \Theta(\sigma, \chi)$$

for all cones $\sigma \in \Sigma$ and all $\chi \in M_{\sigma}$.

Let us compute the Hom-complex for the cone σ_{12} . For $\chi = (a, b)$ and $\psi = (a', b')$, one has

$$P_{\sigma_{12}, \chi} \subset P_{\sigma_{12}, \psi}$$

if and only if

$$a \geq a', \quad b \geq b'.$$

Therefore

$$\text{RHom}(\Theta(\sigma_{12}, (a, b)), \Theta(\sigma_{12}, (a', b'))) \simeq \begin{cases} k, & a \geq a' \text{ and } b \geq b', \\ 0, & \text{otherwise.} \end{cases}$$

On the coherent side, a T -equivariant morphism

$$\mathcal{O}_{U_{\sigma_{12}}}(a, b) \longrightarrow \mathcal{O}_{U_{\sigma_{12}}}(a', b')$$

is given by multiplication by a monomial of weight $(a - a', b - b')$. This monomial is regular on $U_{\sigma_{12}}$ precisely when

$$(a - a', b - b') \in \sigma_{12}^\vee \cap M,$$

which is equivalent to $a \geq a'$ and $b \geq b'$. Hence the coherent Hom-complex agrees with the constructible one:

$$\mathrm{RHom}_T(\Theta'(\sigma_{12}, (a, b)), \Theta'(\sigma_{12}, (a', b'))) \simeq \mathrm{RHom}(\Theta(\sigma_{12}, (a, b)), \Theta(\sigma_{12}, (a', b'))).$$

The same calculation applies to the other maximal cones. For example, for σ_{34} , the corresponding polyhedra are lower-left quadrants, and the inclusion condition becomes $a \leq a'$ and $b \leq b'$. Thus the direction of the inequalities is determined by the chosen cone, exactly as the regularity of monomials is determined by the semigroup $\sigma^\vee \cap M$.

We finally record the image of the usual line bundles. Let $p, q \geq 0$, and consider the divisor

$$D = pD_{\rho_3} + qD_{\rho_4}.$$

The associated polytope is

$$P_D = \{(m_1, m_2) \in M_{\mathbb{R}} \mid 0 \leq m_1 \leq p, 0 \leq m_2 \leq q\}.$$

This is a rectangle. The corresponding line bundle is $\mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1}(p, q)$, and under the correspondence it is represented on the constructible side by the costandard sheaf on this rectangle:

$$\kappa_\Sigma(\mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1}(p, q)) \simeq (i_{P_D})_* \omega_{P_D}.$$

Thus the product structure of $\mathbb{P}^1 \times \mathbb{P}^1$ appears on the constructible side as the rectangular shape of the support. The lattice points of P_D correspond to the usual monomial basis of

$$H^0(\mathbb{P}^1 \times \mathbb{P}^1, \mathcal{O}(p, q)).$$

In this example, the basic dictionary is therefore expressed by rectangles and quadrants: regular equivariant monomial maps correspond to inclusions among rectangular or quadrant-shaped polyhedral sheaves.

8.4 Hirzebruch surfaces $\mathbb{F}_n$

We now consider the Hirzebruch surfaces. This family is useful because it interpolates between the product case and genuinely non-symmetric toric surfaces. The parameter n appears directly in the fan, and hence also in the FLTZ skeleton and in the inclusion conditions for theta polyhedra.

Let $n \geq 0$. The fan Σ_n of $\mathbb{F}_n$ has rays generated by

$$v_1 = (1, 0), \quad v_2 = (0, 1), \quad v_3 = (-1, n), \quad v_4 = (0, -1).$$

The maximal cones are

$$\sigma_{12} = \langle v_1, v_2 \rangle, \quad \sigma_{23} = \langle v_2, v_3 \rangle, \quad \sigma_{34} = \langle v_3, v_4 \rangle, \quad \sigma_{41} = \langle v_4, v_1 \rangle.$$

The associated toric surface is $X_{\Sigma_n} = \mathbb{F}_n$. When $n = 0$, this is $\mathbb{P}^1 \times \mathbb{P}^1$. When $n = 1$, it is the blow-up of $\mathbb{P}^2$ at one torus fixed point.

We first compute the FLTZ skeleton. The zero cone contributes the zero section $M_{\mathbb{R}} \times \{0\}$. The four rays contribute conormal directions along four families of integral affine lines. For $\rho_1 = \mathbb{R}_{\geq 0}v_1$, we get vertical lines

$$\bigcup_{a \in \mathbb{Z}} \{m_1 = a\} \times \mathbb{R}_{\leq 0}v_1.$$

For $\rho_2 = \mathbb{R}_{\geq 0}v_2$, we get horizontal lines

$$\bigcup_{b \in \mathbb{Z}} \{m_2 = b\} \times \mathbb{R}_{\leq 0}v_2.$$

For $\rho_4 = \mathbb{R}_{\geq 0}v_4$, we get the same horizontal base lines, but with the opposite cotangent direction:

$$\bigcup_{b \in \mathbb{Z}} \{m_2 = b\} \times \mathbb{R}_{\geq 0}v_2.$$

The new feature is the ray $\rho_3 = \mathbb{R}_{\geq 0}v_3$. Since

$$\rho_3^\perp = \{(m_1, m_2) \in M_{\mathbb{R}} \mid -m_1 + nm_2 = 0\},$$

this ray contributes the family of affine lines

$$-m_1 + nm_2 = c, \quad c \in \mathbb{Z},$$

with cotangent direction $-\rho_3 = \mathbb{R}_{\geq 0}(1, -n)$. Thus the full FLTZ skeleton is

$$\Lambda_{\Sigma_n} = M_{\mathbb{R}} \times \{0\} \cup \bigcup_{\rho \in \Sigma_n(1)} \bigcup_{\chi \in M_{\rho}} (\chi + \rho^{\perp}) \times (-\rho) \cup \bigcup_{\sigma \in \Sigma_n(2)} \bigcup_{\chi \in M} \{\chi\} \times (-\sigma).$$

Geometrically, the base arrangement consists of the lines

$$m_1 = a, \quad m_2 = b, \quad -m_1 + nm_2 = c,$$

where $a, b, c \in \mathbb{Z}$. For $n = 0$, the last family is again vertical, so one recovers the grid picture of $\mathbb{P}^1 \times \mathbb{P}^1$. For $n > 0$, a new slanted family of lines appears.

We now describe the theta sheaves. The ray generators give costandard sheaves on half-planes:

$$\Theta(\rho_1, a) = (i_{\{m_1 \geq a\}})_* \omega_{\{m_1 \geq a\}}, \quad \Theta(\rho_2, b) = (i_{\{m_2 \geq b\}})_* \omega_{\{m_2 \geq b\}}, \quad \Theta(\rho_4, b) = (i_{\{m_2 \leq b\}})_* \omega_{\{m_2 \leq b\}}.$$

For the slanted ray ρ_3 , if the character class is indexed by $c \in \mathbb{Z}$, then

$$P_{\rho_3, c} = \{(m_1, m_2) \in M_{\mathbb{R}} \mid -m_1 + nm_2 \geq c\},$$

and hence

$$\Theta(\rho_3, c) = (i_{P_{\rho_3, c}})_* \omega_{P_{\rho_3, c}}.$$

Thus the parameter n appears as the slope of the boundary line of this half-plane.

For maximal cones, the theta sheaves are costandard sheaves on translated two-dimensional sectors. If $\chi = (a, b) \in M$, then

$$\begin{aligned} P_{\sigma_{12}, \chi} &= \{m_1 \geq a, m_2 \geq b\}, \\ P_{\sigma_{23}, \chi} &= \{m_2 \geq b, m_1 - nm_2 \leq a - nb\}, \\ P_{\sigma_{34}, \chi} &= \{m_2 \leq b, m_1 - nm_2 \leq a - nb\}, \end{aligned}$$

and

$$P_{\sigma_{41}, \chi} = \{m_1 \geq a, m_2 \leq b\}.$$

The corresponding theta object is

$$\Theta(\sigma, \chi) = (i_{P_{\sigma, \chi}})_* \omega_{P_{\sigma, \chi}}.$$

Compared with the product case, the sectors associated to σ_{23} and σ_{34} have a slanted boundary depending on n .

On the coherent side, each maximal cone gives an affine chart. For example,

$$U_{\sigma_{12}} = \text{Spec } \mathbb{C}[\sigma_{12}^{\vee} \cap M] \cong \mathbb{A}^2,$$

and

$$\Theta'(\sigma_{12}, (a, b)) = (i_{\sigma_{12}})_* \mathcal{O}_{U_{\sigma_{12}}}(a, b).$$

The same construction applies to the other cones, using the corresponding semigroup algebra $\mathbb{C}[\sigma^{\vee} \cap M]$. Under the correspondence,

$$\Theta'(\sigma, \chi) \longmapsto \Theta(\sigma, \chi).$$

Let us compute the Hom-complex for the cone σ_{23} , where the parameter n is visible. For $\chi = (a, b)$ and $\psi = (a', b')$, the inclusion

$$P_{\sigma_{23}, \chi} \subset P_{\sigma_{23}, \psi}$$

holds if and only if

$$b \geq b', \quad a - nb \leq a' - nb'.$$

Equivalently,

$$b - b' \geq 0, \quad -(a - a') + n(b - b') \geq 0.$$

Therefore

$$\text{RHom}(\Theta(\sigma_{23}, (a, b)), \Theta(\sigma_{23}, (a', b'))) \simeq \begin{cases} k, & b \geq b' \text{ and } a - nb \leq a' - nb', \\ 0, & \text{otherwise.} \end{cases}$$

On the coherent side, a T -equivariant morphism between the corresponding theta objects is given by multiplication by a monomial of weight $(a - a', b - b')$. This monomial is regular on $U_{\sigma_{23}}$ precisely when

$$(a - a', b - b') \in \sigma_{23}^{\vee} \cap M.$$

Since

$$\sigma_{23}^\vee = \{u \in M_{\mathbb{R}} \mid \langle u, v_2 \rangle \geq 0, \langle u, v_3 \rangle \geq 0\},$$

this condition is exactly

$$b - b' \geq 0, \quad -(a - a') + n(b - b') \geq 0.$$

Thus the coherent regularity condition agrees with the constructible inclusion condition:

$$\mathrm{RHom}_T(\Theta'(\sigma_{23}, (a, b)), \Theta'(\sigma_{23}, (a', b'))) \simeq \mathrm{RHom}(\Theta(\sigma_{23}, (a, b)), \Theta(\sigma_{23}, (a', b'))).$$

As in the previous examples, one may also look at divisor polytopes to see the shape of familiar line bundle objects. For instance, take

$$D = pD_{\rho_3} + qD_{\rho_4}, \quad p, q \geq 0.$$

Then the associated polytope is

$$P_D = \{(m_1, m_2) \in M_{\mathbb{R}} \mid 0 \leq m_2 \leq q, 0 \leq m_1 \leq p + nm_2\}.$$

This is a trapezoid. The corresponding constructible object is represented by the costandard sheaf on this trapezoid:

$$\kappa_{\Sigma_n}(\mathcal{O}_{\mathbb{F}_n}(D)) \simeq (i_{P_D})_* \omega_{P_D}.$$

Thus the parameter n , which appears in the fan through the ray $(-1, n)$, appears on the constructible side as the slope of the trapezoid and as the slope of the additional conormal directions in the FLTZ skeleton.

The Hirzebruch surface example therefore shows that the CCC is sensitive not only to the number of rays in the fan, but also to their directions. Changing n changes the semigroup conditions for regular monomials on the coherent side, and the same change appears as a change in the inclusion conditions among polyhedral theta sheaves on the constructible side.

8.5 A toric del Pezzo surface

We now consider one toric del Pezzo example. Let X be the blow-up of $\mathbb{P}^2$ at its three torus fixed points. This surface is often denoted by dP_6 . The point of this example is to see how toric blow-ups change the FLTZ skeleton and the corresponding theta sheaves.

Start with the fan of $\mathbb{P}^2$, whose rays are generated by

$$v_1 = (1, 0), \quad v_2 = (0, 1), \quad v_3 = (-1, -1).$$

Blowing up the three torus fixed points means subdividing the three maximal cones by adding the rays

$$v_{12} = v_1 + v_2 = (1, 1), \quad v_{23} = v_2 + v_3 = (-1, 0), \quad v_{31} = v_3 + v_1 = (0, -1).$$

Thus the fan of $X = dP_6$ has six rays. In cyclic order we write them as

$$w_1 = (1, 0), \quad w_2 = (1, 1), \quad w_3 = (0, 1), \quad w_4 = (-1, 0), \quad w_5 = (-1, -1), \quad w_6 = (0, -1).$$

The maximal cones are

$$\langle w_1, w_2 \rangle, \quad \langle w_2, w_3 \rangle, \quad \langle w_3, w_4 \rangle, \quad \langle w_4, w_5 \rangle, \quad \langle w_5, w_6 \rangle, \quad \langle w_6, w_1 \rangle.$$

We first describe the FLTZ skeleton. The zero cone contributes the zero section $M_{\mathbb{R}} \times \{0\}$. The six rays give conormal directions along three families of integral affine lines, with both cotangent directions allowed.

The rays $(1, 0)$ and $(-1, 0)$ give the vertical lines

$$m_1 = a, \quad a \in \mathbb{Z}.$$

The rays $(0, 1)$ and $(0, -1)$ give the horizontal lines

$$m_2 = b, \quad b \in \mathbb{Z}.$$

The rays $(1, 1)$ and $(-1, -1)$ give the diagonal lines

$$m_1 + m_2 = c, \quad c \in \mathbb{Z}.$$

Therefore the one-dimensional part of the FLTZ skeleton contains

$$\bigcup_{a \in \mathbb{Z}} \{m_1 = a\} \times \mathbb{R}_{\leq 0}(1, 0), \quad \bigcup_{a \in \mathbb{Z}} \{m_1 = a\} \times \mathbb{R}_{\geq 0}(1, 0),$$

$$\bigcup_{b \in \mathbb{Z}} \{m_2 = b\} \times \mathbb{R}_{\leq 0}(0, 1), \quad \bigcup_{b \in \mathbb{Z}} \{m_2 = b\} \times \mathbb{R}_{\geq 0}(0, 1),$$

and

$$\bigcup_{c \in \mathbb{Z}} \{m_1 + m_2 = c\} \times \mathbb{R}_{\leq 0}(1, 1), \quad \bigcup_{c \in \mathbb{Z}} \{m_1 + m_2 = c\} \times \mathbb{R}_{\geq 0}(1, 1).$$

The maximal cones add conic directions over lattice points, as prescribed by the general formula

$$\Lambda_\Sigma = M_\mathbb{R} \times \{0\} \cup \bigcup_{\rho \in \Sigma(1)} \bigcup_{\chi \in M_\rho} (\chi + \rho^\perp) \times (-\rho) \cup \bigcup_{\sigma \in \Sigma(2)} \bigcup_{\chi \in M} \{\chi\} \times (-\sigma).$$

We next identify the theta sheaves. The ray theta sheaves are costandard sheaves on the six half-planes

$$m_1 \geq a, \quad m_1 \leq a, \quad m_2 \geq b, \quad m_2 \leq b, \quad m_1 + m_2 \geq c, \quad m_1 + m_2 \leq c.$$

For instance,

$$\Theta(\rho_{(1,1)}, c) = (i_{\{m_1+m_2 \geq c\}})_* \omega_{\{m_1+m_2 \geq c\}},$$

while

$$\Theta(\rho_{(-1,-1)}, c) = (i_{\{m_1+m_2 \leq c\}})_* \omega_{\{m_1+m_2 \leq c\}}.$$

Thus, compared with $\mathbb{P}^2$, the blow-up adds the missing opposite half-plane directions on the constructible side.

For maximal cones, the theta sheaves are costandard sheaves on translated sectors. Consider

$$\sigma = \langle w_1, w_2 \rangle = \langle (1, 0), (1, 1) \rangle.$$

Its dual cone is

$$\sigma^\vee = \{m \in M_\mathbb{R} \mid m_1 \geq 0, m_1 + m_2 \geq 0\}.$$

Hence, for $\chi = (a, b) \in M$,

$$P_{\sigma, \chi} = \chi + \sigma^\vee = \{m_1 \geq a, m_1 + m_2 \geq a + b\}.$$

The corresponding theta sheaf is

$$\Theta(\sigma, \chi) = (i_{P_{\sigma, \chi}})_* \omega_{P_{\sigma, \chi}}.$$

Let $\psi = (a', b')$. Then

$$P_{\sigma, \chi} \subset P_{\sigma, \psi}$$

if and only if

$$a \geq a', \quad a + b \geq a' + b'.$$

Therefore

$$\mathrm{RHom}(\Theta(\sigma, (a, b)), \Theta(\sigma, (a', b'))) \simeq \begin{cases} k, & a \geq a' \text{ and } a + b \geq a' + b', \\ 0, & \text{otherwise.} \end{cases}$$

On the coherent side, a T -equivariant morphism between the corresponding theta objects is given by multiplication by a monomial of weight $(a - a', b - b')$. This monomial is regular on U_σ precisely when

$$(a - a', b - b') \in \sigma^\vee \cap M.$$

Since

$$\sigma^\vee = \{m \in M_\mathbb{R} \mid m_1 \geq 0, m_1 + m_2 \geq 0\},$$

this is equivalent to the same two inequalities

$$a \geq a', \quad a + b \geq a' + b'.$$

Thus

$$\mathrm{RHom}_T(\Theta'(\sigma, (a, b)), \Theta'(\sigma, (a', b'))) \simeq \mathrm{RHom}(\Theta(\sigma, (a, b)), \Theta(\sigma, (a', b'))).$$

This example shows explicitly how the toric blow-up is reflected in the CCC data. On the fan side, blowing up a torus fixed point adds a new ray. On the FLTZ side, this adds a new conormal direction. On the constructible side, it adds new half-plane and sector theta sheaves. The Hom-complex calculation is unchanged in form: it is still governed by inclusion of the corresponding polyhedra, and on the coherent side this is still the regularity condition for the corresponding equivariant monomial map.

9 Generalizations of the Coherent–Constructible Correspondence

The previous sections focused on the coherent–constructible correspondence for smooth toric varieties. In that setting, the input is a fan $\Sigma \subset N_{\mathbb{R}}$, the coherent side is modeled by $\text{Perf}_T(X_{\Sigma})$, and the constructible side is the category of constructible sheaves on $M_{\mathbb{R}}$ whose singular support is contained in the FLTZ skeleton Λ_{Σ} . The key mechanism is the correspondence between coherent theta objects and constructible theta sheaves:

$$\Theta'(\sigma, \chi) \longleftrightarrow \Theta(\sigma, \chi).$$

The Hom-complex comparison is then controlled by inclusions of polyhedra

$$\chi + \sigma^{\vee} \subset \psi + \tau^{\vee}.$$

The purpose of this section is to indicate several directions in which this picture can be generalized. We will not give complete proofs of these generalizations. Instead, we explain how the basic objects of the toric CCC are modified, and how the same organizing principle remains visible.

There are three generalizations that are especially relevant for the present discussion. The first replaces toric varieties by toric Deligne–Mumford stacks. In this case, the fan is replaced by a stacky fan. The constructible side is still controlled by a conic subset of $T^*M_{\mathbb{R}}$, but the lattice data are modified by the stacky structure. This changes the indexing of theta objects and records finite stabilizer information.

The second generalization concerns toric fibrations. Instead of considering a single toric variety over a point, one considers a family whose fibers are toric varieties with fixed fan. The coherent side becomes a category of sheaves on a relative toric space over a base, while the constructible side is no longer just a single sheaf category on $M_{\mathbb{R}}$. It is naturally described in terms of a sheaf, or local system, of categories over the base.

The third generalization concerns Fourier–Mukai transforms. In the usual coherent setting, Fourier–Mukai kernels define functors between derived categories of coherent sheaves. Through the CCC, such functors can be translated into operations on constructible sheaves. This gives a constructible interpretation of certain derived equivalences and birational transformations in toric geometry.

Thus, the guiding question of this chapter is not whether the formula for the CCC remains exactly the same. Rather, it is how each part of the correspondence changes:

$$\begin{aligned} \text{fan} &\rightsquigarrow \text{modified combinatorial input,} \\ \text{Perf}_T(X_{\Sigma}) &\rightsquigarrow \text{coherent category in the new setting,} \\ Sh_{\Lambda_{\Sigma}}(M_{\mathbb{R}}) &\rightsquigarrow \text{constructible or microlocal category with modified support condition.} \end{aligned}$$

In each case, the same basic philosophy remains: algebraic data on the coherent side are translated into constructible sheaves with prescribed microlocal behavior.

9.1 Stacky Fans and Toric Deligne–Mumford Stacks

The first generalization replaces toric varieties by toric Deligne–Mumford stacks. The main change is not the form of the correspondence, but the combinatorial input. Instead of an ordinary fan $\Sigma \subset N_{\mathbb{R}}$, one uses a stacky fan

$$\Sigma = (N, \Sigma, \beta).$$

Here Σ still records the real cone directions, while β records the stacky lattice data along the rays. Thus the constructible side is still controlled by a conic subset of $T^*M_{\mathbb{R}}$, but the indexing of theta objects is modified by the stacky structure.

The basic mechanism remains the same. Coherent theta objects are matched with constructible theta sheaves,

$$\Theta'(\sigma, \chi) \longleftrightarrow \Theta(\sigma, \chi),$$

and Hom-complexes are still controlled by inclusions of polyhedra. What changes is the meaning of the character χ . In the stacky case, the local lattice attached to a cone may be different from the ordinary lattice coming from the coarse toric variety.

We illustrate this by the weighted projective line $\mathbb{P}(1, 2)$. Its coarse moduli space is $\mathbb{P}^1$, but it has a stacky point of order 2. We describe it by the following stacky fan. Let $N = \mathbb{Z}$, and let Σ be the complete fan in $N_{\mathbb{R}} = \mathbb{R}$, with rays

$$\rho_+ = \mathbb{R}_{\geq 0}, \quad \rho_- = \mathbb{R}_{\leq 0}.$$

The stacky vectors are chosen to be

$$b_+ = 1, \quad b_- = -2.$$

Equivalently, the map $\beta : \mathbb{Z}^2 \rightarrow N$ sends the two standard basis vectors to 1 and -2 . The nonprimitive vector -2 records the stacky structure at one torus fixed point.

We now compute the corresponding FLTZ skeleton. The zero cone contributes the zero section

$$\mathbb{R} \times \{0\} \subset T^*\mathbb{R}.$$

For the ray ρ_+ , the local lattice is generated by $b_+ = 1$, so the corresponding character lattice is the ordinary lattice $\mathbb{Z}$. Hence ρ_+ contributes negative conormal rays over the integral points:

$$\bigcup_{a \in \mathbb{Z}} \{a\} \times \mathbb{R}_{\leq 0}.$$

For the ray ρ_- , the local lattice is generated by $b_- = -2$. Its dual lattice is naturally identified with $\frac{1}{2}\mathbb{Z} \subset \mathbb{R}$. Therefore ρ_- contributes positive conormal rays over the half-integral points:

$$\bigcup_{b \in \frac{1}{2}\mathbb{Z}} \{b\} \times \mathbb{R}_{\geq 0}.$$

Thus

$$\Lambda_\Sigma = \mathbb{R} \times \{0\} \cup \bigcup_{a \in \mathbb{Z}} \{a\} \times \mathbb{R}_{\leq 0} \cup \bigcup_{b \in \frac{1}{2}\mathbb{Z}} \{b\} \times \mathbb{R}_{\geq 0}.$$

Compared with the ordinary $\mathbb{P}^1$, the new feature is the appearance of half-integral base points in one conormal direction. This is the constructible trace of the stacky stabilizer of order 2.

The theta sheaves are described in the same way as before, but with the modified indexing set. For $a \in \mathbb{Z}$, the ray ρ_+ gives

$$P_{\rho_+, a} = [a, \infty), \quad \Theta(\rho_+, a) = (i_{[a, \infty)})_* \omega_{[a, \infty)}.$$

For $b \in \frac{1}{2}\mathbb{Z}$, the ray ρ_- gives

$$P_{\rho_-, b} = (-\infty, b], \quad \Theta(\rho_-, b) = (i_{(-\infty, b]})_* \omega_{(-\infty, b]}.$$

Thus the stacky structure does not change the fact that theta sheaves are costandard sheaves on half-lines, but it changes where their endpoints are allowed to lie.

The Hom-complex calculation has the same form as in the ordinary toric case. For $a, a' \in \mathbb{Z}$,

$$\mathrm{RHom}(\Theta(\rho_+, a), \Theta(\rho_+, a')) \simeq \begin{cases} k, & a \geq a', \\ 0, & a < a', \end{cases}$$

because $[a, \infty) \subset [a', \infty)$ if and only if $a \geq a'$. For $b, b' \in \frac{1}{2}\mathbb{Z}$,

$$\mathrm{RHom}(\Theta(\rho_-, b), \Theta(\rho_-, b')) \simeq \begin{cases} k, & b \leq b', \\ 0, & b > b', \end{cases}$$

because $(-\infty, b] \subset (-\infty, b']$ if and only if $b \leq b'$.

This example shows the basic point of the stacky generalization. The formula for the Hom-complex is unchanged; it is still governed by inclusion of polyhedra. What changes is the lattice of allowed translations. In the ordinary $\mathbb{P}^1$ case, both families of endpoints lie in $\mathbb{Z}$. For the weighted projective line $\mathbb{P}(1, 2)$, one family still lies in $\mathbb{Z}$, while the other lies in $\frac{1}{2}\mathbb{Z}$. The fractional positions in the FLTZ skeleton encode the stacky structure.

9.2 Toric Fibrations

The second generalization replaces a single toric variety by a toric fibration. The guiding idea is that the fan still controls the toric directions, while the base of the fibration contributes coefficients to the Hom-complexes.

Let B be a smooth variety, let T be an algebraic torus, and let $P \rightarrow B$ be a principal T -bundle. Given a fan Σ for a toric variety X_Σ , one can form the associated toric fibration

$$X_{\Sigma, P} := P \times_T X_\Sigma \longrightarrow B.$$

The fiber over each point of B is isomorphic to X_Σ . Thus the toric geometry is constant along the base, but the gluing over B may be twisted by the principal T -bundle.

The constructible side should be viewed as a relative version of the ordinary constructible category. The FLTZ skeleton is still determined by the fan Σ , so the microlocal directions in $T^*M_{\mathbb{R}}$ are the same as in the toric variety case. What changes is that the constructible sheaves now carry coefficients coming from the base B . Equivalently, the ordinary condition

$$SS(F) \subset \Lambda_\Sigma$$

is kept, but the values of the sheaves are allowed to remember objects on B , such as perfect complexes or line bundles associated to the principal T -bundle.

The coherent theta objects are also modified in a simple way. For a character $\lambda \in M$, the principal T -bundle $P \rightarrow B$ determines an associated line bundle

$$\mathcal{L}_\lambda := P \times_T \mathbb{C}_\lambda$$

on B , where $\mathbb{C}_\lambda$ denotes the one-dimensional T -representation of weight λ . Thus a monomial of weight λ no longer contributes only a scalar; it contributes the line bundle $\mathcal{L}_\lambda$ on the base. This is the main new feature of the fibration case.

We illustrate this in the simplest example. Let $T = \mathbb{G}_m$, let $P \rightarrow B$ be a principal $\mathbb{G}_m$ -bundle, and take $X_\Sigma = \mathbb{P}^1$. The associated space

$$X_{\Sigma, P} = P \times_{\mathbb{G}_m} \mathbb{P}^1$$

is a $\mathbb{P}^1$ -bundle over B .

The fan is the complete fan in $N_{\mathbb{R}} = \mathbb{R}$, with rays

$$\rho_+ = \mathbb{R}_{\geq 0}, \quad \rho_- = \mathbb{R}_{\leq 0}.$$

Therefore the FLTZ skeleton is the same one as for $\mathbb{P}^1$:

$$\Lambda_\Sigma = \mathbb{R} \times \{0\} \cup \bigcup_{a \in \mathbb{Z}} \{a\} \times \mathbb{R}_{\leq 0} \cup \bigcup_{a \in \mathbb{Z}} \{a\} \times \mathbb{R}_{\geq 0}.$$

Thus the allowed singular directions are unchanged. The difference is that the theta sheaves now carry coefficients from B .

For $a \in \mathbb{Z}$, the ray ρ_+ gives the half-line

$$P_{\rho_+, a} = [a, \infty),$$

and the corresponding constructible theta object is still modeled on

$$(i_{[a, \infty)})_* \omega_{[a, \infty)}.$$

Similarly, the ray ρ_- gives theta objects supported on half-lines of the form $(-\infty, a]$. Hence the polyhedral part of the construction is the same as before.

Now consider the Hom-complex. In the ordinary toric case, one has

$$\mathrm{RHom}(\Theta(\rho_+, a), \Theta(\rho_+, b)) \simeq \begin{cases} k, & a \geq b, \\ 0, & a < b. \end{cases}$$

For the toric fibration, the same inclusion condition remains, but the nonzero term is replaced by cohomology on the base. With the convention above, a morphism from weight a to weight b is represented by a monomial of weight $a - b$. Therefore

$$\mathrm{RHom}_{X_{\Sigma, P}}(\Theta'_P(\rho_+, a), \Theta'_P(\rho_+, b)) \simeq \begin{cases} \mathrm{R}\Gamma(B, \mathcal{L}_{a-b}), & a \geq b, \\ 0, & a < b. \end{cases}$$

The condition $a \geq b$ is exactly the same condition as

$$[a, \infty) \subset [b, \infty).$$

Thus the polyhedral inclusion still determines whether the Hom-complex vanishes, while the base B determines the value of the nonzero Hom-complex.

The same calculation applies to the other ray:

$$\mathrm{RHom}_{X_{\Sigma, P}}(\Theta'_P(\rho_-, a), \Theta'_P(\rho_-, b)) \simeq \begin{cases} \mathrm{R}\Gamma(B, \mathcal{L}_{a-b}), & a \leq b, \\ 0, & a > b. \end{cases}$$

Here the inequality is reversed because

$$(-\infty, a] \subset (-\infty, b]$$

if and only if $a \leq b$.

This example shows the main point of the toric fibration version of the CCC. The fan and the FLTZ skeleton still control the microlocal directions. The theta sheaves are still supported on the same polyhedra in $M_{\mathbb{R}}$. The Hom-complex is still governed by inclusion of these polyhedra. What changes is the coefficient of a nonzero Hom-complex: instead of obtaining k , one obtains the derived global sections of an associated line bundle on the base.

In this sense, the toric fibration version is a relative form of the ordinary toric CCC:

$$\text{polyhedral inclusion} \iff \text{regular monomial direction},$$

but the monomial direction now carries the additional coefficient $\mathcal{L}_\lambda$ coming from the principal torus bundle over B . When $B = \mathrm{Spec} \mathbb{C}$, all associated line bundles have one-dimensional spaces of global sections, and the ordinary toric CCC is recovered.

9.3 Fourier–Mukai Transforms from the Constructible Viewpoint

The third generalization concerns Fourier–Mukai transforms. In the coherent setting, a Fourier–Mukai transform is a way to describe a functor between derived categories by using a kernel on a product. This should be understood as a categorical form of correspondence. Instead of studying only ordinary maps $X \rightarrow Y$, one studies objects on $X \times Y$, which act as generalized relations between X and Y .

Let X and Y be smooth projective varieties, and let

$$K \in \text{Perf}(X \times Y).$$

We regard K as a kernel from X to Y . It defines a functor

$$\Phi_K : \text{Perf}(X) \longrightarrow \text{Perf}(Y)$$

by

$$\Phi_K(E) = \text{R}p_{Y*}(p_X^*E \otimes^{\mathbf{L}} K),$$

where $p_X : X \times Y \rightarrow X$ and $p_Y : X \times Y \rightarrow Y$ are the two projections. Thus the construction has three steps: pull E back to $X \times Y$, tensor it with the kernel K , and then push the result forward to Y .

The reason that K may be viewed as a correspondence is already visible for ordinary morphisms. If $f : X \rightarrow Y$ is a morphism, its graph

$$\Gamma_f \subset X \times Y$$

is a subvariety of the product. The structure sheaf $\mathcal{O}_{\Gamma_f}$ is a kernel, and the associated Fourier–Mukai transform recovers the derived pushforward:

$$\Phi_{\mathcal{O}_{\Gamma_f}}(E) \simeq \text{R}f_*E.$$

Thus an ordinary map gives a special kind of kernel. Fourier–Mukai transforms generalize this by allowing kernels that do not necessarily come from graphs.

Several standard functors are of Fourier–Mukai type. We record the basic examples that will be useful for comparison.

1. The identity functor is given by the diagonal kernel. If $\Delta : X \hookrightarrow X \times X$ is the diagonal embedding, then

$$K = \Delta_*\mathcal{O}_X$$

defines

$$\Phi_K \simeq \text{id}_{\text{Perf}(X)}.$$

Thus the diagonal plays the role of the identity correspondence.

2. Tensoring by a line bundle is also a Fourier–Mukai transform. If L is a line bundle on X , then the kernel

$$K = \Delta_*L \in \text{Perf}(X \times X)$$

defines the functor

$$E \longmapsto E \otimes L.$$

This is the example that we will compute below in the toric setting.

3. Pushforward along a morphism is obtained from the graph kernel. Let $f : X \rightarrow Y$ be a morphism, and let

$$\Gamma_f \subset X \times Y$$

be its graph. Then

$$K = \mathcal{O}_{\Gamma_f} \in \text{Perf}(X \times Y)$$

defines

$$\Phi_K(E) \simeq \text{R}f_*E.$$

Hence an ordinary map $f : X \rightarrow Y$ is a special case of a Fourier–Mukai correspondence.

4. Pullback along a morphism is obtained from the same graph, but with the two factors reversed. If $f : X \rightarrow Y$, then we regard Γ_f as a subvariety of $Y \times X$. The corresponding kernel gives a functor

$$\text{Perf}(Y) \longrightarrow \text{Perf}(X),$$

which is identified, under the usual derived convention, with

$$F \longmapsto \text{L}f^*F.$$

Thus pullback is also represented by a kernel, provided the derived pullback is defined in the chosen category.

5. More generally, a geometric correspondence gives a Fourier–Mukai type functor. Suppose that

$$X \xleftarrow{p} Z \xrightarrow{q} Y$$

is a correspondence and that $F \in \text{Perf}(Z)$. If $(p, q) : Z \rightarrow X \times Y$, then the kernel

$$K = R(p, q)_* F$$

defines a functor of the form

$$E \longmapsto Rq_*(p^* E \otimes^{\mathbb{L}} F),$$

whenever these operations are well-defined. Thus a Fourier–Mukai kernel can be viewed as a categorical correspondence between X and Y .

The word “Fourier” is most visible in the classical case of abelian varieties: there the kernel is the Poincaré line bundle on $A \times \hat{A}$, where $\hat{A}$ is the dual abelian variety. This kernel plays the role of the exponential kernel in the classical Fourier transform. In modern usage, however, the term Fourier–Mukai transform is used more generally for functors defined by kernels.

From the point of view of the coherent–constructible correspondence, the natural question is how such kernel functors appear on the constructible side. Since the CCC identifies coherent theta objects with constructible theta sheaves,

$$\Theta'(\sigma, \chi) \longleftrightarrow \Theta(\sigma, \chi),$$

one can study a Fourier–Mukai type functor by computing its action on theta generators. If the coherent kernel sends theta objects to explicitly described coherent objects, then the CCC translates this action into an operation on polyhedral sheaves.

The example below is different from the toric fibration example of the previous subsection. There, the main point was that nonzero Hom-complexes acquire coefficients from the base. Here we keep a single toric variety and study how a kernel acts as a functor on the theta generators.

Let $X = \mathbb{P}^2$. Its fan has rays generated by

$$v_1 = (1, 0), \quad v_2 = (0, 1), \quad v_3 = (-1, -1),$$

with maximal cones

$$\sigma_{12} = \langle v_1, v_2 \rangle, \quad \sigma_{23} = \langle v_2, v_3 \rangle, \quad \sigma_{31} = \langle v_3, v_1 \rangle.$$

Consider the Fourier–Mukai transform given by tensoring with the equivariant line bundle

$$\mathcal{O}_{\mathbb{P}^2}(1) \cong \mathcal{O}_{\mathbb{P}^2}(D_{\rho_3}).$$

Equivalently, the kernel is

$$K = \Delta_* \mathcal{O}_{\mathbb{P}^2}(1) \in \text{Perf}(\mathbb{P}^2 \times \mathbb{P}^2).$$

To compute its effect on theta generators, we use the local Cartier data of $\mathcal{O}_{\mathbb{P}^2}(1)$. With the above convention, these local characters are

$$m_{\sigma_{12}} = (0, 0), \quad m_{\sigma_{23}} = (1, 0), \quad m_{\sigma_{31}} = (0, 1).$$

Indeed, for D_{ρ_3} , the defining condition is

$$\langle m_\sigma, v_\rho \rangle = -c_\rho$$

for rays $\rho \subset \sigma$, where $c_{\rho_3} = 1$ and $c_{\rho_1} = c_{\rho_2} = 0$.

On the coherent side, tensoring by $\mathcal{O}_{\mathbb{P}^2}(1)$ shifts the local equivariant weight on each affine chart:

$$\Theta'(\sigma, \chi) \longmapsto \Theta'(\sigma, \chi + m_\sigma).$$

Under the CCC, this becomes

$$\Theta(\sigma, \chi) \longmapsto \Theta(\sigma, \chi + m_\sigma).$$

Since

$$\Theta(\sigma, \chi) = (i_{\chi + \sigma^\vee})_* \omega_{\chi + \sigma^\vee},$$

the corresponding constructible operation is the translation

$$\chi + \sigma^\vee \longmapsto \chi + m_\sigma + \sigma^\vee.$$

Let us spell this out for the three maximal cones.

For σ_{12} , one has

$$\sigma_{12}^\vee = \{m_1 \geq 0, m_2 \geq 0\}.$$

Thus

$$P_{\sigma_{12}, (a, b)} = \{m_1 \geq a, m_2 \geq b\}.$$

Since $m_{\sigma_{12}} = (0, 0)$, tensoring by $\mathcal{O}_{\mathbb{P}^2}(1)$ does not shift the theta sheaves on this chart:

$$P_{\sigma_{12},(a,b)} \mapsto P_{\sigma_{12},(a,b)}.$$

For σ_{23} , the dual cone is

$$\sigma_{23}^\vee = \{m_2 \geq 0, m_1 + m_2 \leq 0\}.$$

Hence

$$P_{\sigma_{23},(a,b)} = \{m_2 \geq b, m_1 + m_2 \leq a + b\}.$$

Since $m_{\sigma_{23}} = (1, 0)$, the transform sends this sector to

$$P_{\sigma_{23},(a+1,b)} = \{m_2 \geq b, m_1 + m_2 \leq a + b + 1\}.$$

Thus the corresponding constructible theta sheaf is translated by the local Cartier character $(1, 0)$.

For σ_{31} , the dual cone is

$$\sigma_{31}^\vee = \{m_1 \geq 0, m_1 + m_2 \leq 0\}.$$

Hence

$$P_{\sigma_{31},(a,b)} = \{m_1 \geq a, m_1 + m_2 \leq a + b\}.$$

Since $m_{\sigma_{31}} = (0, 1)$, the transform sends this sector to

$$P_{\sigma_{31},(a,b+1)} = \{m_1 \geq a, m_1 + m_2 \leq a + b + 1\}.$$

This calculation shows how the diagonal kernel

$$\Delta_* \mathcal{O}_{\mathbb{P}^2}(1)$$

appears on the constructible side. It acts on coherent theta generators by shifting their local equivariant weights. Under the CCC, the same operation becomes translation of the polyhedral supports of the corresponding constructible theta sheaves.

The Hom-complex pattern is compatible with this operation. If

$$P_{\sigma,\chi} \subset P_{\sigma,\psi},$$

then after tensoring by $\mathcal{O}_{\mathbb{P}^2}(1)$ the corresponding inclusion becomes

$$P_{\sigma,\chi+m_\sigma} \subset P_{\sigma,\psi+m_\sigma}.$$

Thus the inclusion condition is preserved. Equivalently, on the coherent side, tensoring by a line bundle is an autoequivalence, and the same autoequivalence is seen constructibly as translation of the theta polyhedra.

Therefore this subsection illustrates a different use of the CCC from the toric fibration case. A toric fibration changes the values of the Hom-complexes by introducing coefficients from a base. A Fourier–Mukai kernel, by contrast, acts as a functor on the CCC category itself. In the example above, the kernel $\Delta_* \mathcal{O}_{\mathbb{P}^2}(1)$ acts by translating the polyhedral supports determined by the local Cartier data of the line bundle.

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